

The Supply-Side Effects of Healthcare Privatization: Evidence from Puerto Rico*

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Abstract

We provide, to our knowledge, the first causal evidence on privatizing an entire public healthcare delivery system. We study Puerto Rico's 1993 health reform, which privatized all three functions of a government-run system at once: it transferred facilities to private ownership, replaced salaried public employment of providers with private capitated contracts, and placed the public insurance under private managed-care administration. Because the reform rolled out across all 78 counties between 1994 and 2000, we estimate its effects with the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021). Findings indicate that the healthcare sector contracted, where per thousand residents, employment fell 13.0 percent and Medicare-certified hospital beds 7.8 percent relative to their pre-reform means. The healthcare labor market adjustment was seen in employment levels rather than wages. Total population coverage contracted even as formal insurance among enrollees rose. Patient health did not deteriorate on average, and infant mortality fell by 13 to 27 percent of its pre-reform mean in the more urban, populous, and healthcare-intensive counties. A back-of-the-envelope calculation estimates the freed real resources at about \$844 million in 2025 dollars over the four post-reform years under our most conservative estimate, enough to make net welfare positive before health gains.

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[‡][The Appendix and Online Appendix are available here](#)

1 Introduction

Health spending has risen steadily as a share of GDP across OECD countries, averaging 9.3 percent in 2024, roughly double its 1970 level, and reaching 17.2 percent in the United States (Organisation for Economic Co-operation and Development, 2025).¹ Governments finance most of this spending. Across the OECD, government and compulsory schemes cover close to three-quarters of health costs, and in the United States Medicare, Medicaid, and the Children’s Health Insurance Program together accounted for about 7 percent of GDP in 2024 (Centers for Medicare and Medicaid Services, 2025a; Medicaid and CHIP Payment and Access Commission, 2026). To contain these costs, governments reform the delivery of care along three margins, facility ownership (switched to private), provider payment (switched to capitated), and coverage (switched to more restrictive). A large literature establishes the effect of each in isolation. Private ownership raises provider profitability (Duggan, Gupta, Jackson, & Templeton, 2023), capitation strengthens the incentive to economize on medical inputs (Glied, 2000; Gruber, 2017), and coverage changes shift the demand for care (Finkelstein et al., 2012). Because these reforms change the incentives providers face, their effects reach beyond public spending to the allocation of workers, patients, and resources. Governments continue to enact them at scale, with capitated managed care now administering more than half of U.S. Medicaid spending (Kaiser Family Foundation, 2025) and the 2018 VA MISSION Act moving the government’s Veterans Health Administration delivery system towards private providers (Congressional Research Service, 2025a). What such reforms do to the health workforce and to patients is therefore a pressing question. The existing literature cannot speak to a whole-system reorganization, since it comes almost entirely from policies that move a single margin within an otherwise intact market. To fill this literature gap, we study a reform that changed all three margins at once, Puerto Rico’s 1993 privatization of its public health system, and ask what privatizing an entire delivery system does to its labor

¹On the CMS National Health Expenditure Accounts basis, which additionally counts the capital investment the OECD current-expenditure measure excludes, U.S. national health expenditures were 18.0 percent of GDP in 2024 (Centers for Medicare and Medicaid Services, 2025a).

market and its patients.

Before the reform, the Commonwealth ran a centralized system of direct public provision, owning the facilities, employing salaried providers, and administering universal coverage. Between 1994 and 2000, across five staggered cohorts of the island’s 78 counties, the reform privatized all three functions: it sold or leased its facilities to private providers, replaced provider salaries with private-sector capitated contracts, and transferred coverage to an income-based public insurance program, capped at 200 percent of the Puerto Rico poverty line and privately administered by Medicaid managed-care organizations (MCOs) (Commonwealth of Puerto Rico, 1993). All three were used as cost containment tools by the government, and together they replaced public delivery with the private delivery of publicly financed care.

What the privatization literature lacks, and what Puerto Rico supplies, is the privatization of direct public provision: a system the government owned, staffed, and delivered. The settings it studies keep a public or private alternative in place (Chan, Card, & Taylor, 2023; Goodair & Reeves, 2022), privatize a few facilities inside an intact market (Duggan et al., 2023; Tiemann & Schreyögg, 2012; Tvaliashvili, Hellowell, Roubal, Zoidze, & Clarke, 2025), or change how providers are paid or how many patients are covered while providers remain private (Aizer, Currie, & Moretti, 2007; Duggan, 2004; Duggan, Gruber, & Vabson, 2018; Finkelstein et al., 2012), so a surrounding market absorbs the displaced patients and workers and contaminates any control group. Puerto Rico instead converted its whole system at once; its coverage change was itself a contraction, an income-based insurance that narrowed effective coverage through take-up frictions and coverage gaps (Finkelstein, Hendren, & Shepard, 2019; Finkelstein & Notowidigdo, 2019). With no untreated public provider or beneficiary left on the island, no cross-border patient or provider flows, and federal Medicaid financing that is capped (Congressional Research Service, 2025b), the market adjustments play out inside a closed market and the design is free of the spillovers and funding incentives that complicate mainland studies.

Estimating the reform’s effect across the 78 counties with the not-yet-treated staggered difference-in-differences estimator of Callaway and Sant’Anna (2021), applied to administrative and vital-statistics data, we find a dual contraction of supply: health care employment fell 13.0 percent and hospital beds fell 7.8 percent. Quality did not fall so much as reallocate. We find an improvement (or lack of deterioration) for the monitorable, profitable conditions that capitation and private ownership reward, such as cardiovascular care, and decreasing for the less contractible, less profitable ones, such as obstetrics, where maternal mortality rose. This confirms that the reorganization toward more profitable services following privatization, documented by Duggan et al. (2023) for hospitals, also extends to the broader healthcare delivery system.

This paper makes three contributions. First, it provides the first causal evidence, to our knowledge, on privatizing direct public provision, which tests the property-rights theory of Hart, Shleifer, and Vishny (1997). Second, it measures how Medicaid managed care performs against direct public delivery, adding the delivery margin that studies comparing insured and uninsured populations hold fixed (Finkelstein, 2007; Finkelstein, Hendren, & Shepard, 2019). Third, it informs the growing practice of contracting publicly financed care to private providers, from the United State’s VA’s community-care program to the United Kingdom’s NHS clinical outsourcing to New Zealand’s elective procurement (Congressional Budget Office, 2021; Health New Zealand (Te Whatu Ora), 2024; Institute for Fiscal Studies, 2019), by showing how a completed version of that shift performs against the public delivery it replaced.

The paper proceeds as follows. Section 2 describes the reform and the pre-reform system, Section 3 the theoretical framework, Section 4 the data, Section 5 the empirical strategy, Section 6 the results, and Section 7 the robustness analysis. Section 8 concludes.

2 Setting

In plain terms, the reform changed who owned the clinics, how their staff were paid, and how patients reached care. Before 1993, a low-income resident of Puerto Rico could walk into a government clinic, be seen by a salaried public physician, and be treated at no charge, and the public system stood open to everyone as a fallback. After the reform, that resident enrolled through an income-tested health card, was assigned to the single private managed-care plan for their region, and reached care through a primary-care gatekeeper for a small copayment, while the clinics and hospitals themselves passed to private owners paid a fixed amount per enrollee rather than a salary. The Commonwealth continued to pay for the care of the low-income population but no longer delivered it. This section describes the system that changed and the terms of that change.

2.1 The Pre-Reform Public Delivery System

As of the mid-1950s, the Commonwealth of Puerto Rico delivered healthcare through an integrated public system, built on the regionalized principles of Arbona and Ramirez de Arellano (1978), with universal access at the point of service, a four-tier referral hierarchy, and direct public employment of clinical staff. The base tier comprised 78 Centers of Diagnostic and Treatment (CDTs), one per county, providing primary, prenatal, dental, and pharmaceutical care through salaried government clinicians; above them sat five Area Hospitals, six Regional Hospitals, and the supra-tertiary Puerto Rico Medical Center in San Juan. The system was the dominant source of care: in 1990 it operated about 40 percent of the Commonwealth's hospital beds and served roughly 1.8 million residents, about 52 percent of the population (Instituto de Administración y Política de Salud, 1995).

The reform confronted a public delivery system bound by a hard budget constraint. By 1990 total health spending had reached \$2.5 billion, about 11 percent of gross domestic product (World Health Organization & Pan American Health Organization, 1998), yet federal

Medicaid funds arrived as a capped allotment, so the Commonwealth financed most public provision from its own budget (Kaiser Family Foundation, 2024; Medicaid and CHIP Payment and Access Commission, 2020). According to Commonwealth of Puerto Rico (1993), this constraint pressed hardest on the public sector because a two-tier structure had sorted residents by ability to pay: privately insured residents used private providers, while the public Department of Health remained the residual provider for a poorer, sicker, and adversely selected population. A fixed budget could not match that population’s need for capital and medical technology, and the under-resourcing was visible in outcomes, with contemporaneous assessments documenting quantity and quality shortfalls relative to private providers and higher recorded mortality among the publicly served (World Health Organization & Pan American Health Organization, 1998). The enabling legislation attributed the decline in public-sector quality to inadequate budgets, the rising cost of medical technology, bureaucratic centralization, and political interference (Commonwealth of Puerto Rico, 1993). The reform’s stated aim was to extend quality care to the low-income population while reducing cost, by contracting private insurers and providers under capitation (Commonwealth of Puerto Rico, 1993).

2.2 The 1993 Reform

The Puerto Rico Health Insurance Administration Act (Act 72 of September 7, 1993) created a single public purchaser of coverage, the Administración de Seguros de Salud (ASES), and restructured delivery along all three margins at once (Commonwealth of Puerto Rico, 1993; World Health Organization & Pan American Health Organization, 1998).² Ownership moved from public to private: the 78 CDTs (one per county) and the 11 Area and Regional Hospitals

²Eligibility extended to residents with household incomes below 200 percent of the Puerto Rico poverty threshold, alongside the existing Medicaid-eligible population; the benefits package covered inpatient and outpatient hospital care, physician and ambulatory services, pharmacy, dental, mental-health and substance-abuse treatment, and preventive services, without exclusions for pre-existing conditions. Cost-sharing was \$1 to \$4 per visit and \$0.50 to \$3 per prescription, scaled by income (Centers for Medicare and Medicaid Services, 2000, 2001, 2003; Commonwealth of Puerto Rico, 1993; Instituto de Administración y Política de Salud, 1995; World Health Organization & Pan American Health Organization, 1998).

were sold or placed under long-term lease to private providers, while the Commonwealth retained the supra-tertiary Medical Center and Cardiovascular Center (Commonwealth of Puerto Rico, 1993; Instituto de Administración y Política de Salud, 1995). Provider payment moved from Department of Health salaries to capitation, paid by ASES-contracted insurers at rates negotiated bilaterally with providers (Commonwealth of Puerto Rico, 1993; World Health Organization & Pan American Health Organization, 1998). Access moved from open entitlement at public facilities to enrollment in a single managed-care network (designated by county of residence) with primary-care gatekeeping and income-scaled cost-sharing. The public system had been an open outside option for the privately insured and the main one for the uninsured, so its transfer to private providers removed that fallback.³ Financing details and post-2002 policy changes, including the successor programs, are reported in Appendix 10.

ASES divided the Commonwealth into eight administrative health regions and awarded each to a single managed-care organization through competitive bidding, creating regional monopolies, and rolled the regions out between February 1994 and July 2000 (Brugueras Fabre, 2002; World Health Organization & Pan American Health Organization, 1998) (see Figure 1 panel (a) for a map of the cohorts across all counties). The number of distinct managed-care organizations participating island-wide grew from two in 1995 to seven by 2005, with two present throughout the period and several smaller plans entering and exiting. Mental-health services were carved out to separate behavioral-health organizations beginning in 2002.⁴ The order ran from the sparsely populated and outer-island regions to the capital of San Juan, so treatment timing is correlated with county urbanization, the identification concern we address in Section 5. We group the eight transition dates into five annual cohorts by year of transition: 1994 (10 counties), 1995 (12), 1996 (30), 1998 (25), and 2000 (San Juan

³Since we lack data on the size of the new provider networks, we cannot say whether enrollees gained access to more providers or fewer.

⁴Counts of participating organizations by year (two in 1995; four in 1996; three in 1997; four in 1998 and 1999; three in 2000; four in 2001; six in 2002; five in 2003; and seven in 2004 and 2005) are compiled from the CMS Medicaid managed-care enrollment reports for 1995–1999 (Centers for Medicare and Medicaid Services, 1995, 1996, 1997, 1998, 1999) and the CMS national summaries of state Medicaid managed-care programs for 2000–2005 (Centers for Medicare and Medicaid Services, 2000, 2001, 2002, 2003, 2004, 2005); the 1995 report records two participating organizations without naming them.

alone). Because San Juan holds the largest concentration of healthcare establishments and employment, we report its influence separately and show that excluding it attenuates the supply-side magnitude (Section 7).

As the regions implemented the policy, Medicaid managed-care enrollment rose from 130,392 residents in 1994 to 1,795,067 by 2000, about 47 percent of the population and roughly 96 percent of the eligible (Administración de Seguros de Salud de Puerto Rico, 2026), while the program’s outlay scaled from essentially zero to about \$1.9 billion in 2025 dollars (about 5 percent of the Commonwealth’s consolidated budget),⁵ most of it premium pass-through to the contracting managed-care organizations rather than direct public provision (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1999).⁶ During this period, the government shifted its health spending from operating public facilities toward purchasing private premiums, while patients faced rising costs for medical services that outpaced healthcare workers’ average wages (see Figure 1, Panel (c)⁷ and Panel (d)⁸).

⁵All shares here are computed in nominal fiscal-year-2000 dollars. Puerto Rico’s consolidated budget that year was \$19.9 billion, of which the General Fund was \$7.0 billion (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1999). ASES’s \$1.0 billion was about 5 percent of the consolidated budget; public health overall (ASES plus the \$0.9 billion Department of Health, some \$1.9 billion) was about 10 percent.

⁶In nominal terms the outlay was about \$1.0 billion in fiscal year 2000; the text deflates it to 2025 dollars with the U.S. CPI-U. The implied program cost averaged about \$557 per enrollee per year (\$46 per member per month) in nominal fiscal-year-2000 dollars, a descriptive average of appropriations over enrollment rather than a negotiated capitation rate.

⁷Panel (c) is in constant 2025 dollars, deflated with the U.S. CPI-U. Public delivery is the cost of the government-operated agencies (the Assistance Fund for Health Services, AFASS; the Medical Center, ASEM; and the Cardiovascular Center), reported on a financial-statements basis that runs below the operating-budget totals used elsewhere; private delivery is the capitated premiums the government pays the managed-care organizations through ASES, not out-of-pocket or private-insurance spending; and total government health spending is their sum. The household medical-care share on the right axis is a national-accounts share of total personal consumption, not out-of-pocket spending, shown for comparison only and not comparable to the dollar levels. Sources: the Commonwealth budget volumes (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1997, 1999, 2001; Estado Libre Asociado de Puerto Rico, Oficina de Presupuesto y Gerencia, 1989, 1993; Gobierno de Puerto Rico, 2006) for spending and Junta de Planificación de Puerto Rico (2024) for the consumption share.

⁸The real wage per healthcare worker in Panel (d) is the annual QCEW wage bill, total quarterly wages summed over the four calendar quarters, divided by annual average employment for NAICS 621–623 in Puerto Rico, in constant 2025 dollars (U.S. CPI-U); it is a covered-wage measure that excludes self-employed physician income and managed-care-organization margins. Fiscal years run July–June and are labeled by their ending year, whereas the QCEW wage is a calendar-year quantity, so the two are aligned by year label. Sources: Junta de Planificación de Puerto Rico (2024) (Table 5) and the U.S. Bureau of Labor Statistics Quarterly Census of Employment and Wages (U.S. Bureau of Labor Statistics, 2025).

3 Theoretical Framework

We organize the analysis around the property-rights theory of the firm (Hart et al., 1997; Shleifer, 1998). Ownership and provider payment jointly determine the provider’s incentive to reduce costs when quality is imperfectly contractible, while publicly financed insurance determines the demand for care the provider faces. Under private ownership the provider retains its cost savings and cannot be penalized for reducing quality on dimensions no contract specifies, so its incentive to economize strengthens and its incentive to sustain noncontractible quality weakens. Capitation places the same incentive on the payment margin: the provider is residual claimant on the fixed per-member payment (Aizer et al., 2007; Duggan, 2004; Marton, Yelowitz, & Talbert, 2014). Because the employment outcome is derived labor demand and labor is the largest variable input in care delivery, both incentives fall first on clinical staffing, so each margin predicts a reduction in per-capita healthcare employment, $\Delta L^{\text{ownership}} \leq 0$ and $\Delta L^{\text{capitation}} \leq 0$, the latter unless the capitated rate rises to offset it.

Publicly financed insurance enters through demand. Replacing universal public provision with an income-based, application-based entitlement raises formal coverage among those who enroll but reaches a smaller share of the population: take-up is incomplete and lowest among the neediest (Finkelstein & Notowidigdo, 2019), and residents just above the eligibility threshold lose the entitlement without gaining coverage (Finkelstein, Hendren, & Shepard, 2019). Primary-care gatekeeping and cost-sharing further curb utilization among the enrolled. The net is a contraction in demand-relevant coverage that lowers the demand for care and, with it, derived labor demand, $\Delta L^{\text{access}} \leq 0$. Whether patient health moves with these three predictions is not similarly determined.

Unlike labor, which the reform reduces directly, health reflects the noncontractible quality q , on which the property-rights logic admits both signs: Hart et al. (1997) hold that private provision is generally cheaper but may raise or lower quality, since private ownership strengthens the incentive to economize on cost and the incentive to invest in quality alike. Which incentive prevails depends on the pre-reform baseline. Where the public system

was adequately resourced the cost-reducing incentive dominates and $\Delta q \leq 0$; where it was capacity-constrained the quality-investment incentive fills a binding gap and $\Delta q > 0$. Which regime holds is an empirical question, and its answer depends on how the inputs the reform removed are valued, a point we return to in the welfare analysis (Section 6.5).

The three margins act on the same derived labor demand, so their labor predictions combine into the reform’s net effect on employment,

$$\Delta L^{\text{reform}} = \underbrace{\Delta L^{\text{ownership}}}_{\leq 0} + \underbrace{\Delta L^{\text{capitation}}}_{\leq 0} + \underbrace{\Delta L^{\text{access}}}_{\leq 0}. \quad (1)$$

This combined effect is what the design identifies, because the margins interact rather than add. The incentive to economize under capitation is stronger where the provider also owns the facility. The demand margin interacts with both: capitated private providers have reason to select toward more profitable patients, so the coverage contraction and supply-side selection compound. Duggan et al. (2023) document this selection in single-hospital privatizations, where market-level admissions of low-income patients fell even with coverage held fixed. Because the reform moved all three margins at once and they bear on different populations and markets, this design identifies ΔL^{reform} rather than the contribution of any one.⁹

4 Data

The analysis combines six data sources, all used at the county level and at annual frequency within 1990–2001.¹⁰ We measure the local healthcare sector in two ways. Employment comes from the U.S. Bureau of Labor Statistics Quarterly Census of Employment and Wages (QCEW) (U.S. Bureau of Labor Statistics, 2025), which reports county-quarter employment,

⁹Appendix 10 gives the formal derivations of the ownership, capitation, and access components, the quality corollary, and the mobility margin.

¹⁰Except for the CMS Provider of Services files (Centers for Medicare and Medicaid Services, 2025b), which are used only from 1991.

wages, and establishment counts by NAICS code; the healthcare sector is the union of ambulatory care (NAICS 621), hospitals (622), and nursing and residential care (623), and we report NAICS 621 separately because it maps most directly onto the privatized county clinic network. Because the QCEW counts establishments of every ownership type, transferring facilities from public to private does not by itself move workers into or out of the measure, since a change in employment is a change in the number of healthcare jobs, not a reclassification across ownership categories. Hospital capacity comes from the Centers for Medicare and Medicaid Services Provider of Services (POS) files (Centers for Medicare and Medicaid Services, 2025b), aggregated to the county-year. The two measures see different parts of the system: the POS files identify only Medicare-certified providers and so exclude the 78 CDTs, whose primary care did not require certification. The privatized clinic layer that was the reform’s main target therefore shows up in QCEW employment, not in the bed count, which captures only the hospital tier.

The Puerto Rico Department of Health’s Vital Statistics reports (Puerto Rico Department of Health, 2025) supply the patient-health outcomes (the crude birth rate, infant mortality, the low-birth-weight share, and the government and private hospital-birth shares), along with county population and the crude net migration rate; the Department issued these only as unstructured annual PDFs, which we digitized, reconciled across years, and assembled into the structured county-year panel used here. The worker-margin outcomes (labor-force participation, the employment-to-population ratio, and unemployment) come from the monthly county labor-force survey of the Puerto Rico Department of Labor and Human Resources (Puerto Rico Department of Labor and Human Resources, 2025). Two further sources support the robustness checks: the National Center for Health Statistics National Vital Statistics System (NVSS) natality files (National Center for Health Statistics, 2025) corroborate the county birth outcomes with individual records, and the Behavioral Risk Factor Surveillance System (BRFSS) (Centers for Disease Control and Prevention, 2025) supplies the individual adult coverage reports we use to test the access first stage.

Appendix 10 gives the construction details.

The Vital Statistics population series (the Commonwealth’s official population count) is the denominator for all per-thousand-resident outcomes. Per-capita scaling aligns the labor, supply, and health outcomes with the literature (Hackmann, Heining, Klimke, Polyakova, & Seibert, 2025; Wallace, 2023). The balanced QCEW healthcare panel covers 59 counties (76% of the total) and the balanced CMS panel 40 (51% of the total), while the Vital Statistics and labor-force panels cover 78 (100% of the total); the smaller QCEW panel reflects the Bureau of Labor Statistics’ suppression of county-industry cells with few establishments or little employment, which tilts the balanced healthcare sample toward counties with more concentrated healthcare markets. The supply-side estimates are therefore identified on a sample that already over-weights densely populated counties, so their concentration in the metropolitan areas should be read as consistent with both heterogeneity and sample composition, a caveat we address in Section 7.

Table 1 reports summary statistics, and Table 2 the cohort-specific pre-reform (1990–1993) means of population, public-system intensity, healthcare and placebo-industry employment, labor-market activity, demographic flows, and patient-health covariates, with one-way ANOVA p -values for the joint null of equal cohort means. The cohorts differ systematically in urban intensity, reflecting the rural-to-urban rollout.

5 Empirical Strategy

The empirical analysis exploits the staggered rollout of the reform across five cohorts of counties ($\mathcal{G} = \{1994, 1995, 1996, 1998, 2000\}$) to identify the average effect of treatment on the treated (ATT). Let $Y_{i,t}$ denote an outcome measured for county i in year t , let $G_i \in \mathcal{G}$ denote the treatment cohort of county i , and let $D_{i,t} = \mathbf{1}\{t \geq G_i\}$ be the post-treatment indicator. Because every county in Puerto Rico is treated by 2000, the analysis has no never-treated comparison group; identification relies instead on *not-yet-treated* coun-

ties, units in later cohorts $g' > g$ not yet treated by year t . The estimand of interest is the group-time average treatment effect $\text{ATT}(g, t) = \mathbb{E}[Y_{i,t}(g) - Y_{i,t}(\infty) \mid G_i = g]$ for $t \geq g$ (Callaway & Sant’Anna, 2021), where $Y_{i,t}(g)$ and $Y_{i,t}(\infty)$ are the potential outcomes under treatment-in-cohort- g and no-treatment, respectively. Identification rests on three assumptions: (i) conditional pre-treatment parallel trends for the not-yet-treated comparison group, $\mathbb{E}[Y_{i,t}(\infty) - Y_{i,t-1}(\infty) \mid G_i = g] = \mathbb{E}[Y_{i,t}(\infty) - Y_{i,t-1}(\infty) \mid G_i > t]$; (ii) no anticipation, $Y_{i,t}(g) = Y_{i,t}(\infty)$ for $t < g$; and (iii) staggered, irreversible adoption. Under (i)–(iii), $\text{ATT}(g, t)$ is identified by the doubly robust difference-in-differences estimator

$$\widehat{\text{ATT}}(g, t) = \mathbb{E}[Y_{i,t} - Y_{i,g-1} \mid G_i = g] - \mathbb{E}[Y_{i,t} - Y_{i,g-1} \mid G_i > t]. \quad (2)$$

The regression tables report the simple weighted average of $\widehat{\text{ATT}}(g, t)$ across cohorts and post-treatment periods, with weights proportional to the number of post-treatment county-year observations; the event-study figures report the dynamic aggregation $\widehat{\theta}_\ell$ for $\ell \in \{-4, \dots, +4\}$ with the reference period $\ell = -1$. Inference uses the multiplier bootstrap of Callaway and Sant’Anna (2021) clustered at the county level with 1,000 replications.

Because the estimand and the parallel-trends assumption are estimator-specific, Section 7 re-estimates every main outcome with three alternatives: the canonical OLS two-way fixed-effects estimator (TWFE), the extended two-way fixed-effects (ETWFE) estimator of Wooldridge (2025), and the imputation estimator of Borusyak, Jaravel, and Spiess (2024) (BJS). The rollout also produces cohort imbalance on observable pre-treatment levels, since earlier-treated counties are systematically more rural than later-treated ones (Table 2); under Callaway and Sant’Anna (2021) (C&S) such imbalance does not threaten identification so long as the not-yet-treated comparison group satisfies assumption (i). We assess that assumption directly through the event-study pre-trend coefficients reported alongside each main result in Section 6.

Three features of the setting bear on the identifying assumptions, and Section 7 tests

each. The first is geography, and it works in the design’s favor. Because Puerto Rico is an island, we assume it generated no spillovers to healthcare markets beyond it, removing the cross-border confound that complicates privatization studies set in the continental United States. The residual concern is internal: labor and patient flows across treated and not-yet-treated counties could contaminate the comparison group that assumption (i) relies on. We assess it by re-estimating on counties that share no border with a county treated in a different year. The second is the IRS Section 936 corporate-tax phase-out, a manufacturing tax incentive whose withdrawal began in 1996, contemporaneous with the rollout. It is an unlikely confound for the healthcare estimates. Manufacturing under §936 was concentrated in a small set of industrial counties distinct from the healthcare centers, above all San Juan, and manufacturing and healthcare share no input-output linkage, so a shock to the one does not transmit mechanically to the other. Any island-wide component of the phase-out fell on treated and not-yet-treated counties alike, so the not-yet-treated comparison that assumption (i) uses differences it out. We nonetheless condition the main estimate on pre-reform manufacturing exposure directly (Section 7.5); the estimate attenuates but keeps its sign, and we read the attenuation as manufacturing exposure standing in for the urban, industrial markets where the healthcare contraction itself concentrated rather than as §936 driving the result. The third is the leverage of San Juan as the sole 2000 cohort. With no county treated later, its group-time effect has no not-yet-treated comparison and is weakly identified, so we report its influence separately.

6 Results

6.1 First Stage and Supply Contraction

The reform moved births almost one-for-one from government to private hospitals, the best available proxy in our data for how ownership changed the inpatient delivery system. Table 3 and its event study, Figure 1 panel (b), report the estimates. The government-hospital share

fell by 14.79 percentage points ($p < 0.01$), or 23.6 percent of its 62.55 percent pre-reform mean, and the private-hospital share rose by an offsetting 15.19 percentage points ($p < 0.01$). The two estimates are statistically indistinguishable in absolute magnitude and opposite in sign, so the reorganization of births is essentially a one-for-one substitution from public to private facilities.¹¹ The event study confirms that the divergence in trends begins at the start of the policy.

Coincident with the ownership transition, per-capita healthcare employment fell. Table 4 reports the labor-market estimates for the healthcare aggregate (NAICS 621+622+623) and, separately, for ambulatory primary care (NAICS 621), the subsector most directly associated with the privatized county clinic network. Healthcare employment per thousand residents fell by 0.511 employees, 13.0 percent of its pre-treatment mean ($p < 0.01$), with a larger decline of 17.3 percent in ambulatory primary care ($p < 0.01$). Establishments per thousand fell by 6.6 percent in the aggregate ($p < 0.01$) and by 4.6 percent in NAICS 621 ($p < 0.10$), so the contraction operated on both the extensive margin (net establishment exit) and the intensive margin (fewer employees per establishment).

This per-capita decline is a population-weighted average and does not mean the sector shrank everywhere: in a $\log(1 + x)$ specification the same outcomes grow, a distributional split we interpret in Section 7.6 (Appendix Table A.7). The contraction concentrated in the dense, more concentrated urban markets and spared the rural periphery, the pattern Duggan et al. (2023) find in highly concentrated markets.

The adjustment ran through the quantity of labor, not its price. Table 4 shows employment and establishment counts falling while compensation per worker stayed flat, the wage-per-employee estimate statistically indistinguishable from zero in both panels. This confirms the prediction of Section 3, where ownership and capitation give providers reason to cut their largest variable input rather than wages, and it matches the input economizing found in single-hospital privatization (Duggan et al., 2023) and under Medicaid capitation

¹¹The two shares are not exact complements because a small fraction of births, typically under one percent, is recorded as non-hospital or with unspecified ownership.

(Duggan, 2004). Those results, like the payer-side evidence that managed care lowers spending through lower utilization and product substitution rather than price (Curto, Einav, Finkelstein, Levin, & Bhattacharya, 2019; Dranove, Ody, & Starc, 2021), describe a private insurer economizing on the care it pays for; ours describes the delivery system itself shrinking the inputs it operates.

The labor-market event study confirms flat pre-trends. Figure 2 reports it across six panels, for employment, establishments, and the wage bill per employee under both industry definitions. The pre-treatment coefficients are statistically flat through $\ell = -1$, satisfying the parallel-trends assumption, and the post-reform contraction in employment and establishments begins within the first year while the wage per employee stays flat throughout.

The establishment decline also thinned provider competition within each region, a market-structure margin the framework in Section 3 does not model and one a single-facility privatization cannot identify. Whether that thinning lowered quality, or the replacement of an under-resourced public provider with better-equipped private provision raised it, is the question the patient-health estimates take up (Section 6.4).

Physical hospital capacity contracted as well, on the narrower CMS panel of forty Medicare-certified counties. These estimates appear alongside the patient-health outcomes in Table 5, Columns 3–4: Medicare-certified hospital beds per thousand fell by 7.8 percent ($p < 0.01$) and Medicare-certified services per thousand by 4.9 percent ($p < 0.05$). The contraction therefore spans both layers of the delivery system, the ambulatory clinic network (QCEW employment and establishments) and the inpatient hospital infrastructure (CMS beds and services). Because births occur in the hospital tier, the first-stage substitution measures the inpatient transition directly, while the largest employment reduction falls on the ambulatory clinic tier the bed count cannot see; Act 72 moved both tiers together (Section 2.2), so the hospital birth-share proxies a system-wide ownership transition rather than an inpatient-only one. The bed and service event studies appear in Figure 3, Panels (e)–(f), where the

post-treatment contraction opens by $\ell = +2$.¹²

6.2 The Coverage Margin

Formal insurance among the low-income rose, though the survey evidence for it is underpowered. Across age, education, and employment subgroups, self-reported insurance coverage in the BRFSS rises by 3 to 10 percentage points, estimated under the Wooldridge (2025) extended two-way fixed-effects estimator because the post-1996 BRFSS window admits only the later cohorts.¹³ An administrative series corroborates this survey-based result. The Medicaid managed-care enrollment introduced in Section 2.2 tracks the cohort rollout (Appendix Figure A.3) and is the program-level counterpart to the survey-measured coverage gain (Administración de Seguros de Salud de Puerto Rico, 2026).¹⁴ ASES’s own tabulation of insured against eligible population shows that take-up was nonetheless incomplete, as it typically is for income-based programs (Finkelstein & Notowidigdo, 2019): the insured reached 86 percent of the eligible in 1995–96 and 93 to 98 percent thereafter, leaving 2 to 14 percentage points of eligible residents uninsured (Administración de Seguros de Salud de Puerto Rico, 2026). The Puerto Rico reform enrolled the low-income group itself, so formal insurance for those who took up widened, unlike the facility-level privatizations of Duggan et al. (2023), where privatization instead cut low-income admissions market-wide. Access for the general population is another matter. We do not observe the uncovered share directly, but replacing universal entitlement at public facilities with an income-based card, together

¹²The supply magnitudes reported here are simple-aggregate ATTs, averaged over all post-treatment periods. The event studies in Figures 2 and 3 show the per-capita contraction deepening with time since treatment, so the longest-horizon coefficients exceed these averages, with correspondingly wide confidence intervals. We report the simple-aggregate ATT and use it in the welfare accounting (Section 6.5) as the conservative choice.

¹³Online Appendix Tables OA.14–OA.19 report the six BRFSS subgroup estimates; the “has health plan” point estimates range from +2.8 percentage points (ages 65 and older, where near-universal Medicare leaves little room to expand) to +10.5 percentage points (ages 45–64), and are significant in five of the six subgroups. Appendix 10 discusses the subgroup pattern and the single-cohort-exposure limitation that requires the ETWFE estimator on this panel.

¹⁴The enrollment counts are the ASES administrative tabulation (Administración de Seguros de Salud de Puerto Rico, 2026), which supersedes the budget-figure read plotted in the appendix figure; the series corroborates the access margin descriptively and does not identify a separate effect.

with the incomplete take-up above and the loss of the public outside option for the near-poor who did not qualify (Finkelstein, Hendren, & Shepard, 2019), implies that coverage for the population at large narrowed even as formal enrollment rose among the low-income who took up. The reform thus withdrew a universal public entitlement and replaced it with income-based private coverage reaching only part of the population: formal insurance widened for enrollees while access contracted for the population at large.

6.3 Worker-Side Adjustment Margins

The contraction shows up in labor-force participation and out-migration, not in unemployment or total population. Table 6 reports the margins along which displaced healthcare workers and the broader local labor market might adjust. The labor-force-participation rate fell by 1.11 percentage points, 3.5 percent of its 31.895 percent pre-treatment mean ($p < 0.01$), while the BLS-standard unemployment rate is statistically indistinguishable from zero against a pre-reform mean of 17.770 percent.¹⁵ The crude net migration rate moves from a pre-treatment mean of +1.06 residents per thousand to an estimated post-treatment -14.65 per thousand ($p < 0.05$), and the crude birth rate falls 3.1 percent (-0.564 per thousand, $p < 0.05$). Total county population, by contrast, does not fall; the population-level estimate is +5.1 percent ($p < 0.01$).¹⁶ These estimates rule out a population exodus as the driver of the contraction, since treated counties grew even as participation and the crude birth rate fell. We do not observe age composition, so we infer no specific demographic recomposition: participation and fertility fell, but not because the population contracted. The net-migration and crude-birth-rate event studies appear in Figure 3, Panels (c)–(d), the migration trend changing within two years of treatment and the birth-rate trajectory

¹⁵The unemployment rate is reported in Online Appendix Table OA.1, alongside the labor-force-participation and employment-to-population rates; Appendix 10 reports the participation event study, which satisfies parallel trends and begins declining at the policy rollout.

¹⁶Online Appendix Tables OA.20 and OA.21 report the population estimate in levels and logs and an explicit demographic-balancing check. The population-level and net-migration estimates do not arithmetically reconcile, because the migration rate is a residual of three separately measured flows (see Equation 4), more sensitive to compounding measurement error than the level outcome, and because the Callaway and Sant’Anna (2021) aggregation weights differ across panels of different length.

bending downward from $\ell = 0$.

These margins also bound what we can say about the displaced workers themselves. Because unemployment did not rise, they were not absorbed into local joblessness; participation and net migration both fell instead. County aggregates cannot follow individuals, so we cannot separate labor-force exit, migration off the island, and reallocation into other local sectors. Two features make selective provider emigration the most plausible margin. First, the population grew, so any out-migration was selective rather than broad. Second, Puerto Rico’s clinicians hold U.S. credentials that require no re-licensing on the mainland (Appendix 10), which lowers the cost of leaving for precisely the workers the reform displaced. The negative net-migration estimate is consistent with this selective outflow.

The net migration rate is the one primary outcome whose sign is sensitive to estimator choice, so we read its magnitude as suggestive (Section 7). The participation result, on the same 78-county panel and the behavioral counterpart of the migration hypothesis, is robust across all four estimators.

6.4 Patient Health and Heterogeneity

The short-horizon effect on patient health is null on average. Table 5, Columns 1–2, reports the county-aggregate Vital Statistics outcomes whose response horizons fall within the four-year post-reform window: the infant mortality rate fell by 1.205 deaths per thousand live births (9.5 percent of its pre-mean, statistically insignificant) and the low-birth-weight share rose by 0.06 percentage points (statistically insignificant), and both event studies in Figure 3, Panels (a)–(b), are flat in the pre- and post-treatment windows.¹⁷

The average null hides offsetting movements across causes of death, and which movements the four-year window can register depends on their response horizon. Acute-care deaths are readable within the window through case-fatality, whereas chronic-disease mortality responds too slowly to appear; the Hispanic health paradox (Fernández, García-Pérez,

¹⁷Individual-level NVSS specifications on birth weight, prenatal-visit count, trimester of prenatal-care initiation, and Apgar scores are similarly null (see Online Appendix Table OA.22).

& Orozco-Alemán, 2023), under which this population shows lower mortality than its risk profile predicts, implies that measured health can deteriorate well before mortality registers it. Read within these limits, the cause-specific estimates in Appendix Table A.9 sort into two groups that proxy different dimensions of care.

The first group collects deaths that timely preventive and chronic-disease care can avert, and on the cardiovascular margins it points to improvement: heart-disease mortality falls 6.1 percent ($p < 0.05$) and cerebrovascular mortality 11.0 percent ($p < 0.05$), while diabetes and pneumonia or influenza mortality are unchanged. Maternal mortality moves the other way, rising 73 percent of its small pre-reform base (+0.002 deaths per thousand residents, $p < 0.10$); because a maternal death usually reflects prenatal risk left unmanaged rather than delivery-room failure, this is consistent with acute delivery capacity holding while the preventive care that depends on sustained patient contact decreased. The maternal margin is the population-health counterpart of the selective retrenchment Duggan et al. (2023) document, in which private providers withdraw from the least profitable, low-income service lines.

The second group, deaths from accidents and homicide, proxies emergency and trauma care, where survival turns on acute-care access that often does not cross county lines. Both rise, accident mortality by 40 percent (statistically insignificant) and homicide mortality by 75 percent ($p < 0.01$), consistent with the contraction of beds and facilities lengthening the path to emergency care and with the reform leaving little capacity elsewhere to absorb the displaced demand: each region was served by a single managed-care network, the limiting case of the concentrated-market condition under which Duggan et al. (2023) find that competing hospitals do not take up the slack and market-level low-income admissions fall. We read the homicide estimate cautiously, since its recorded count also reflects the incidence of violence, which rose across Puerto Rico through the 1990s for reasons unrelated to the health system.

Across the within-window outcomes, the decomposition suggests offsetting movements

rather than a broad deterioration: improvement on the cardiovascular margins, no average change in infant mortality, and deterioration on the obstetric margin and, more tentatively, on emergency-care access.

The reform bit hardest in the urban, higher-capacity counties, not in the rural periphery that relied most on public facilities beforehand. Table 7 partitions the 78 counties at the median pre-reform government-hospital birth share: the first-stage substitution is larger in the below-median group (-15.49 percentage points, $p < 0.01$) than in the above-median group ($+2.41$ percentage points, statistically insignificant). Baseline reliance is the wrong intuition for where the reform bit, because pre-reform reliance on government facilities was highest in the rural, poorer counties with the fewest healthcare facilities. That high-reliance half is largely the rural periphery, where the same facilities went on delivering most births, so measured substitution was small. The realized ownership transition instead concentrated in the urban, more populous, higher-capacity, richer, and lower-unemployment counties. Partitioning on pre-reform population density, population, market concentration (healthcare employment), the poverty rate, or the unemployment rate, the government-hospital share falls by 24 to 31 percent in the urban, larger, higher-capacity, richer, and lower-unemployment half and is flat in the other (see Appendix Table A.6 and Online Appendix Tables OA.9, OA.10, OA.23, and OA.24).

The infant-mortality rate follows the same pattern. In the halves where the ownership transition took hold, infant mortality falls by 13 to 27 percent of its pre-treatment mean, statistically significant across the density, population, market concentration (healthcare-employment), poverty, unemployment, and commuting partitions; in the halves where the substitution was muted it does not improve, and in the smallest counties it rises, a marginally significant 28 percent.¹⁸ The gain concentrates in the urban, lower-poverty, higher-capacity

¹⁸For the half in which the ownership transition concentrated, the infant-mortality ATTs in deaths per thousand live births are -2.03 (urban, $p < 0.05$), -2.56 (larger population, $p < 0.01$), -2.63 (higher healthcare employment, $p < 0.01$), -1.62 (lower poverty, $p < 0.10$), -2.62 (lower unemployment, $p < 0.05$), and -3.12 (workplace-center, $p < 0.01$); the government-hospital substitution in these halves ranges from -14.2 to -18.2 percentage points. In the opposite halves the infant-mortality estimates are positive and insignificant, except the smallest-population half at $+3.66$ ($p < 0.10$). Appendix Table A.6 and Online

counties with the private infrastructure to absorb the transfer, the markets Duggan et al. (2023) find most resilient to an ownership change, and it does not concentrate in the poorest counties where the coverage gain was largest, so it does not trace the access margin. Because these counties differ on many dimensions at once and the subsamples are small, we read the pattern as consistent with quality gains from the transition rather than a clean isolation of which margin produced them. The improvement is hard to square with the uniform quality deterioration Hart et al. (1997) predicts, but it falls within that theory’s boundary condition set out in Section 3 rather than refuting it: an under-resourced public baseline on which physical capacity binds, so the private quality-investment incentive dominates the cost-reducing one and quality rises. The supporting heterogeneity event studies, by baseline poverty rate and by pre-reform government-hospital exposure, appear in Figures 4 and 5, each splitting the panel at the median and tracing employment, infant mortality, and net migration for the two halves.

The result runs opposite to Duggan et al. (2023), who find that U.S. hospital privatization raised elderly mortality; the difference is largely one of population and horizon, since their effects fall on a horizon our four-year perinatal window does not reach, and among the adult causes we can read at a short horizon maternal mortality moves up as their elderly mortality does while cardiovascular mortality moves down. It also differs from the payer-only managed care that Aizer et al. (2007) find raised neonatal mortality: the reform bundled a coverage gain with delivery privatization, and the infant gains concentrate where private infrastructure absorbed the transfer. The payer-side evidence otherwise fits the no-harm reading, since Duggan et al. (2018) find that Medicare Advantage restricted hospital access with no measurable mortality effect, though that comparison sets a private payer against a public one rather than private against public delivery and so only bounds our interpretation. Because mortality is a coarse proxy for quality, a caveat Duggan et al. (2018) attach to their own no-harm finding, our null cannot rule out movements in dimensions of care that

Appendix Tables OA.9, OA.10, OA.23, OA.24, and OA.11 report all six partitions.

mortality does not capture. A competing measurement concern does not bind here, since the recorded-count inflation from managed-care risk-adjustment and quality-reporting incentives that Chorniy, Currie, and Sonchak (2018) document does not reach our physical-capacity and vital-statistics outcomes.

The realized improvement also speaks to the market-structure route left open in Section 6.1. The establishment decline thinned provider competition, which in settings such as the English NHS, where hospitals compete on quality at administered prices, has been linked to lower mortality and better management (Bloom, Propper, Seiler, & Van Reenen, 2015; Cooper, Gibbons, Jones, & McGuire, 2011; Gaynor, Moreno-Serra, & Propper, 2013), so thinner competition would, other things equal, work against quality. That infant mortality nonetheless fell where the transition concentrated is consistent with the better-equipped private provision of the boundary condition outweighing any competition loss, though the two cannot be separated here.

6.5 Welfare Bounds

We close by combining the estimated supply and health effects into a back-of-the-envelope cost-benefit calculation, following Duggan et al. (2023). We report it in constant 2025 dollars throughout, as elsewhere in the paper.¹⁹ Read for sign and order of magnitude rather than as a point estimate, it treats the beds and workers the reform freed as a welfare gain under two assumptions: that care quality is maintained, which the data support only as the absence of a detectable full-panel change, and that the saving reflects freed real resources rather than reduced provider rents.

We define net welfare as the avoided cost of the healthcare inputs the reform no longer

¹⁹The calculation is carried out in real 1982–1984 dollars to match the price base of Duggan et al. (2023) and converted to constant 2025 dollars with the U.S. CPI-U (all items, U.S. city average, 1982–1984 = 100; 2025 annual average ≈ 321.9 , a factor of about 3.22). The 1982–1984-dollar values are reported in the footnotes that follow and in Appendix Table A.4, which gives every cell in both price bases.

purchased plus the value of any averted infant deaths:

$$W = \underbrace{\frac{PT}{1000} \left(|\hat{\beta}_{\text{bed}}| c_b + |\hat{\beta}_{\text{emp}}| \bar{w} \right)}_{\text{avoided cost}} + \underbrace{v \frac{|\hat{\beta}_{\text{IMR}}|}{1000} B_H T}_{\text{value of averted infant deaths}}. \quad (3)$$

We depart from Duggan et al. (2023) in three respects. First, the cost saving is the avoided cost of the inputs the delivery system saved: the per-thousand reductions of $\hat{\beta}_{\text{bed}} = -0.265$ beds (see Table 5) and $\hat{\beta}_{\text{emp}} = -0.511$ workers (see Table 4),²⁰ scaled to the island population over the four post-reform years. We measure these physical inputs directly, because the reform privatized an entire delivery system whose budget we do not observe at the provider level.²¹ A bed-year is priced at its implied total operating cost c_b from the CMS cost reports (Centers for Medicare and Medicaid Services, 2023a) and a worker-year at the 1990–1993 mean healthcare wage $\bar{w}$. Second, the health term is a benefit rather than a cost, since infant mortality fell, whereas they find added deaths among the elderly. Third, because the full-panel infant-mortality reduction is statistically insignificant, we value it only as an upper bound, applying it over the births in the high-substitution half B_H of Section 6.4, one-half the island total, at the U.S. EPA reference value of \$10 million per statistical life (U.S. Environmental Protection Agency, 2010; Viscusi & Aldy, 2003).²²

Under the zero-health choice the bound rests on the avoided cost alone and is \$967 million under the employment estimate (\$69 per resident per year). Conditioning the employment reduction on the contemporaneous manufacturing-tax phase-out (Section 7.5) attenuates it to -0.224 per thousand and lowers the avoided cost to \$844 million (\$60 per resident per year), our most conservative figure. Adding the value of the averted infant deaths, about

²⁰These are the simple average treatment effects on the treated, not the larger end-of-horizon event-study coefficients; pricing the period-average rather than the peak effect keeps the avoided-cost term conservative.

²¹This avoided cost is the value of the physical inputs the system no longer used, not a reduction in public outlay. Aggregate public health spending in fact rose over this period, as purchasing through ASES scaled past \$1 billion in nominal terms (Appendix 10), so the welfare improvement is a fall in real-input use under maintained quality rather than a smaller public budget, a pattern consistent with Duggan (2004), who finds that mandatory Medicaid managed care raised government spending.

²²This \$10 million reference value is in 2024 dollars, equivalent to about \$10.3 million in the constant 2025 dollars used throughout the paper and about \$3.19 million in the 1982–1984 dollars in which the calculation is carried out.

\$3.1 billion, raises net welfare to \$4.0 billion under the main supply estimate and \$3.9 billion under the §936-conditional one (\$289 and \$280 per resident per year).²³

We take the zero-health cells as our main estimate: they are positive, rest only on the avoided cost, and do not depend on the patient-health margin in either direction. Among them we prefer the §936-conditional figure, which nets out the manufacturing-tax phase-out. The health-valued cells are an upper bound, for three reasons. First, the infant-mortality benefit they monetize is insignificant on the full panel and recovered only in subgroups. Second, they price that benefit while ignoring the adult cause-specific effects, which move both ways: the favorable cardiovascular declines against the adverse maternal increase, the clearest identified short-horizon harm. Third, they apply an adult statistical-life value to infant deaths, understating the per-death value, since an averted infant death returns far more life-years than the elderly deaths Duggan et al. (2023) price; valuing the health term in life-years rather than in lives would only raise our benefit. Four further margins lie beyond what we can quantify, and none changes the sign of our estimate: mortality beyond the four-year window, the value of the coverage gain to enrollees (which low-income willingness-to-pay places below its cost (Finkelstein, Hendren, & Shepard, 2019)), the share of the saving that reflects reduced provider rents rather than freed real resources, since much healthcare spending is such a transfer (Finkelstein, Hendren, & Luttmer, 2019), and whether the freed labor and capacity were redeployed productively, since the welfare value of freed inputs depends on frictions in the input market (Hackmann et al., 2025).

Read in these terms, our result stands opposite the single-hospital case. The same ac-

²³In 1982–1984 dollars these are an avoided cost of \$300.8 million (main supply estimate) and \$262.5 million (§936-conditional), an averted-infant-death benefit of \$958.7 million valued at a \$3.19 million statistical life, and net welfare of \$1,259.5 million (main) and \$1,221.2 million (§936-conditional); Appendix Table A.4 reports every cell in both price bases. This upper bound values only the averted infant deaths, and on a footing favorable to the reform: the infant benefit is recovered in the high-substitution subgroup, while on the full panel the infant-mortality effect is statistically indistinguishable from zero. The contemporaneous maternal-mortality increase (see Appendix Table A.9) is a full-panel estimate working the other way; we flag it as an offset rather than netting it in, because the marginally significant maternal coefficient off a very small base is not on the subgroup footing used for the infant term. The asymmetry of footings is itself why we read these health-valued cells as an upper bound: on a single consistent footing the monetized health term would be much smaller, leaving the avoided cost as the binding source of the welfare gain.

counting leads Duggan et al. (2023) to judge that case socially inefficient, since the roughly \$2.7 million their average privatization saves the local government comes with about 3.6 additional elderly deaths, some \$0.8 million of saving per death lost and an order of magnitude below the \$10 million value of a statistical life. Our positive sign instead rests on the inputs the system saved standing against a health margin that improved on its dominant component, so the bound clears zero even before any health benefit is counted.

7 Robustness

7.1 Estimator class

We compare the main estimates against three alternatives: the canonical two-way fixed-effects estimator, the extended two-way fixed-effects estimator of Wooldridge (2025), and the imputation estimator of Borusyak et al. (2024).²⁴ TWFE serves as a diagnostic. When treatment effects vary across cohorts and over time, as they do here since the rollout moves from rural to more urban counties, the TWFE coefficient is a negatively weighted average of the underlying effects and can reverse sign. C&S, ETWFE, and BJS are built to avoid that negative weighting, and we rely on them for the main estimates. The three agree in sign and magnitude on the first-stage transition, the healthcare-employment contraction, the labor-force-participation decline, and the hospital-bed contraction, while TWFE produces the expected sign reversals precisely on the outcomes where heterogeneity is most pronounced (healthcare employment, infant mortality, and the low-birth-weight share).²⁵ The crude net migration rate is the one outcome on which the four estimators disagree. TWFE, ETWFE,

²⁴See Appendix Figure A.1 for the comparison event studies across all estimators.

²⁵Appendix Table A.1 reports the four-estimator point estimates and bootstrap standard errors for the six primary outcomes, and the full TWFE specification (county and year fixed effects with event-time indicators, reference period omitted, standard errors clustered at the county level). Across estimators, the first-stage government-hospital-share estimate is -12.9 percent (TWFE), -26.4 percent (ETWFE), -23.6 percent (C&S), and -18.0 percent (BJS) of pre-mean, all $p < 0.01$; HC employment per thousand is -25.7 (ETWFE), -13.0 (C&S), and -16.7 (BJS), all $p < 0.01$; LFP per population is -3.6 , -3.5 , and -2.2 percent, all $p < 0.01$; hospital beds per thousand is -13.1 , -7.8 , and -5.6 percent, all $p < 0.01$.

and BJS return insignificant estimates of mixed signs (-5.69 , -1.53 , and $+4.51$ per thousand), whereas the C&S estimator yields a significant -14.65 per thousand ($p < 0.05$), the estimate on which the migration discussion in Section 6.3 rests. We therefore read it as suggestive of net out-migration, relying on its sign rather than its magnitude.

7.2 Excluding San Juan

Dropping San Juan, the sole 2000-cohort county and roughly 11 percent of the Commonwealth’s population, attenuates the supply-side and labor-market estimates while leaving the first stage intact. The government-hospital-share estimate moves from -14.79 to -8.22 percentage points (both $p < 0.01$),²⁶ healthcare employment from -13.0 to -5.3 percent of pre-mean and the hospital-bed estimate from -7.8 to $+3.0$ percent, the latter two no longer significant. The attenuation is mechanical, since the capital holds the largest concentration of healthcare workforce and Medicare-certified capacity, so dropping it strips out most of the units of observation through which a per-capita contraction is identified. Where that concentration sits is itself informative. The capital is not only the largest market but the lowest-poverty and highest-capacity one, the setting in which Duggan et al. (2023) find privatization least likely to reduce access, because the surrounding market can absorb the change. The reform’s largest input reduction therefore fell where access was most insulated, and the infant-mortality improvement concentrates in these same counties (Section 6.4); the San Juan-inclusive magnitude is the effect where the public system was largest and most readily absorbed by the private sector.

7.3 Within-island spillovers

The not-yet-treated comparison requires that already-treated neighbors not affect comparison counties (Section 5). We test this with two sample restrictions: a non-border subset

²⁶Online Appendix Table OA.3 reports the first-stage and patient-health side-by-side specifications; Appendix Table A.2 reports the QCEW labor-market and CMS hospital-capacity specifications; Online Appendix Table OA.4 reports the ownership-stratified CMS robustness.

(where we drop counties that share a boundary with a different-cohort treated unit), and a low-commuting-inflow subset (where we drop the upper half of the workplace in-commuting distribution (U.S. Census Bureau, 2000a)).²⁷ The first-stage transition is robust under both (-12.14 percentage points on the non-border subset, -7.18 on the low-commuting subset, both $p < 0.05$), as are the patient-health outcomes. The labor-market estimates do not transfer to the non-border restriction at its sample size of 27 QCEW counties, where health-care employment, the wage bill per employee, and establishments are each indistinguishable from zero, and the hospital-bed and services coefficients keep the sign of the main estimates but lose precision. We read this as an effect of power loss rather than evidence against the main estimates.

7.4 Placebo industries

These placebo tests check whether a contemporaneous shock, such as the §936 corporate-tax phase-out beginning in 1996, contaminates the estimate. We re-run the design on industries with no exposure to the privatized delivery system: NAICS 11 (agriculture), NAICS 48–49 (transportation), and NAICS 71 (arts); an effect there would signal a common shock rather than a reform effect. NAICS 31–33 manufacturing is excluded *a priori*, since the §936 phase-out shocks it during the window, making it endogenous to the same policy as healthcare rather than orthogonal.²⁸ The evidence is mixed. Agriculture is null on absolute-headcount and per-capita employment and establishments, and its per-capita wage-bill decline does not survive raw-level normalization, consistent with population redistribution rather than an agriculture-specific shock. Transportation contracts in both per-capita and raw levels, beyond what redistribution can produce, consistent with input-output linkage to the phasing-out manufacturing supply chain. Arts is underpowered ($n = 9$). Because the transportation

²⁷Appendix Table A.5 and Online Appendix Tables OA.6, OA.5, OA.7, and OA.8 report the corresponding specifications. The non-border restriction retains 27 of 78 counties on the QCEW panel, 19 of 40 on the CMS panel, and 39 of 78 on the Vital Statistics panel. The low-commuting-inflow restriction retains 64 of 78 on the Vital Statistics panel and 34 of 40 on the CMS panel.

²⁸Online Appendix Tables OA.12 and OA.13 report the per-thousand-resident and raw-level placebo specifications.

result runs through a known confounder, we turn to a direct manufacturing control next.

7.5 Manufacturing control

The transportation placebo failure motivates a direct test of whether the main contraction is identified in isolation from the §936 phase-out. We condition the main healthcare-employment estimate on a pre-reform manufacturing-exposure covariate, the 1990–1993 mean of NAICS 31–33 employment per thousand; using a strictly pre-1996 baseline rather than the contemporaneous series ensures the covariate is not itself shocked by the policy whose contamination the test nets out.²⁹ The C&S estimate attenuates from -14.7 percent of pre-mean (no control, $p < 0.01$) to -6.8 percent (with the covariate, statistically insignificant), remaining negative and not driven by sample restriction. The result is therefore neither pure §936 contamination, since the conditional estimate is negative rather than zero, nor identified in isolation at full magnitude. We interpret this as the pre-reform manufacturing exposure absorbing part of the cross-sectional heterogeneity in the contraction, a correlation that may reflect §936 contamination but may also reflect that high-manufacturing counties are urban and industrial and so overlap with the dimensions on which the healthcare contraction concentrates (Section 6.4).

7.6 Per-capita normalization

The estimates show a -13.0 percent contraction in healthcare employment per thousand and a roughly $+19.0$ percent growth in raw-level $\log(1 + y)$ employment. We interpret this as the local healthcare sector growing in absolute terms but more slowly than the local population.³⁰ We interpret this difference as distributional. The $\log(1 + y)$ raw-level measure

²⁹Appendix Table A.3 reports the conditioning specification. The sample is the 54 counties with non-missing 1990–1993 baseline manufacturing data. The C&S no-control estimate on this sample is -14.7 percent of pre-mean, economically equivalent to the 78-county main result of -13.0 percent reported in Table 4, so the sample restriction itself does not drive the conditioning result.

³⁰Appendix Table A.7 reports per-1,000 and $\log(1 + y)$ specifications for QCEW healthcare employment, establishments, and the wage bill side by side; Online Appendix Table OA.2 reports the parallel specification on $\log(1 + y)$ per-1,000 normalizations and $\log(1 + y)$ raw-level outcomes.

weights counties roughly equally and is pulled upward by the many low-baseline rural counties that expanded proportionally, whereas the per-1,000 measure is population-weighted and dominated by the high-baseline urban centers, which contracted both absolutely and per resident. The three measures answer different questions: the raw-level estimate is the change in the absolute job count, the per-1,000 estimate the change in jobs per resident, and the $\log(1 + y)$ estimate the change in proportional terms. The +19.0 percent is the $\log(1 + y)$ figure, driven up by the small, low-baseline rural counties where adding a few jobs is a large percentage gain, whereas the -13.0 percent is the per-1,000 figure, driven down by the large urban centers, where most residents live and where jobs per resident fell.

8 Conclusion

Puerto Rico's 1993 health reform sold or leased its public healthcare facilities to private providers, paid private managed-care organizations prospective capitation, and replaced open public entitlement with an income-based card. We find that it reduced healthcare resources per capita: healthcare employment, ambulatory primary-care employment, the establishment count, and Medicare-certified hospital beds and services all contracted relative to population. The adjustment ran through labor-force participation rather than unemployment, alongside declines in the crude birth rate and net migration; the displaced workers were not absorbed into local unemployment, and the participation and migration declines are consistent with labor-force exit. Patient health did not broadly deteriorate within the four-year window, and it improved where the ownership transition took hold, with infant mortality falling 13 to 27 percent of its pre-mean in the urban, more populous, higher-capacity, and richer counties.

The main finding is that privatizing an entire public delivery system can deliver less measured healthcare per capita without a proportional loss in measured patient quality. The estimates do not vindicate the Hart et al. (1997) quality-deterioration corollary in this setting, since infant mortality fell where the reform had the most bite. Nor do they show

the supply expansion a net coverage gain would predict: the local healthcare labor market contracted rather than expanded under the bundled treatment of capitation, single-network confinement, and ownership transfer, and the access margin was itself a coverage contraction for the population at large once the universal public fallback was withdrawn and the income-based card reached only part of the eligible (Finkelstein, Hendren, & Shepard, 2019; Finkelstein & Notowidigdo, 2019).

They instead support the claim that an ownership-and-payment change that constrains the per-enrollee budget can reduce per-capita inputs without proportional welfare losses when the pre-reform public system was capacity-constrained, so the private quality-investment incentive dominates the cost-reducing one (Hart et al., 1997; Propper, 2018). The budget record corroborates this, since the Commonwealth dissolved its public delivery agencies and routed financing through Medicaid managed care, whose outlay scaled to about \$1.9 billion in 2025 dollars by fiscal year 2000 even as the per-capita inputs to the local healthcare sector fell (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1999, 2002).

A back-of-the-envelope bound puts the short-horizon welfare gain at about \$844 million in 2025 dollars over the four-year window on the avoided cost alone, the binding source of welfare improvement even when patient health is valued at zero (Section 6.5); an illustrative monetization of the subgroup infant-mortality improvement would raise the ceiling to about \$4.0 billion but rests on the favorable subgroup and an offsetting maternal increase, so we do not rely on it. The same cost-benefit accounting that Duggan et al. (2023) use to find single-hospital privatization socially inefficient yields a positive sign here, because the inputs the system gave up stood against a patient-health margin that improved rather than deteriorated.

For the privatization debates, the contribution is the counterfactual itself. The reforms that inspire those debates each move a partial bundle of the three margins Puerto Rico moved together: the Department of Veterans Affairs MISSION Act added outside-option access without transferring ownership, the English internal market introduced competition without

capitating public delivery, and risk-based Medicaid managed care capitates payment without transferring facilities. Puerto Rico is the only case in which the bundle is complete and the delivery system itself changes ownership, so it speaks to the public-versus-private-delivery question the payer-side and single-facility literatures cannot reach. Those settings change who pays while providers stay private (Chorniy et al., 2018; Curto et al., 2019; Dranove et al., 2021; Duggan et al., 2018; Wallace, 2023), or compare public and private provision within a system where both exist (Chan et al., 2023), or convert a single facility within an intact market (Duggan et al., 2023); none observes the transition from public delivery system to private. Puerto Rico did so through procurement to a single plan per region, the market design Cuesta and Tebaldi (2025) show can raise welfare relative to regulated competition by removing plan-level cream-skimming, though it leaves provider-level selection intact.

The pattern this case establishes should carry to other settings. Contracting an entire publicly operated delivery network to private providers under capitation should contract per-capita inputs to the local healthcare sector. By the same logic, reversing that design and moving a system like Medicaid managed care toward direct government delivery would add per-capita inputs, though our estimates come from privatization rather than its reverse and give no assurance that patient mortality would fall in step. The comparison here runs in favor of the private model on resource use, because Puerto Rico’s public system carried more inputs per resident without delivering better measured outcomes; whether welfare improves depends on the pre-reform system’s resourcing, the local provider-mobility option, and the incidence of the contraction.

9 Tables

Table 1: County Descriptive Statistics by Year

	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001
Panel A: QCEW — Healthcare Labor Market [†]												
Employment per 1k pop.	2.86 (4.74)	3.28 (5.15)	3.44 (5.05)	3.61 (5.23)	3.90 (5.41)	4.03 (5.33)	4.54 (5.56)	5.02 (5.85)	5.52 (6.49)	5.92 (6.76)	6.22 (7.16)	7.67 (9.94)
Establishments per 1k pop.	0.62 (0.54)	0.66 (0.54)	0.73 (0.57)	0.73 (0.58)	0.75 (0.57)	0.82 (0.60)	0.89 (0.62)	0.93 (0.64)	0.99 (0.65)	1.07 (0.67)	1.11 (0.68)	1.13 (0.69)
Wage bill per 1k pop. (\$)	5,405 (10,809)	6,558 (12,859)	7,059 (13,280)	7,454 (13,985)	7,966 (13,962)	8,478 (14,211)	9,490 (15,151)	10,651 (16,101)	11,852 (18,296)	12,772 (19,145)	13,439 (20,646)	16,981 (26,925)
<i>Counties</i>	59	59	59	59	59	59	59	59	59	59	59	59
<i>Treated Counties</i>	—	—	—	—	5	16	38	38	58	58	59	59
Panel B: CMS — Provider of Services [‡]												
<i>Total Providers</i>	—	202	222	242	248	254	259	262	265	268	274	275
<i>Government</i>	—	42	42	48	48	48	44	44	44	45	45	43
<i>Private</i>	—	160	180	194	200	206	215	218	221	223	229	232
Providers per 1k pop.	— (—)	0.08 (0.05)	0.09 (0.06)	0.09 (0.06)	0.09 (0.07)	0.10 (0.07)	0.10 (0.07)	0.10 (0.07)	0.10 (0.07)	0.10 (0.07)	0.10 (0.07)	0.10 (0.07)
Beds per 1k pop.	— (—)	3.17 (3.08)	3.15 (3.06)	3.21 (3.06)	3.18 (3.04)	3.18 (3.09)	3.16 (3.14)	3.10 (3.09)	3.08 (3.07)	3.18 (3.28)	3.20 (3.26)	3.21 (3.26)
Services per 1k pop.	— (—)	0.46 (0.47)	0.48 (0.49)	0.48 (0.49)	0.49 (0.49)	0.49 (0.50)	0.49 (0.49)	0.48 (0.47)	0.47 (0.47)	0.48 (0.47)	0.48 (0.47)	0.49 (0.51)
FTE per 1k pop.	— (—)	9.04 (8.34)	9.17 (8.44)	9.54 (8.45)	12.85 (19.54)	12.54 (19.46)	12.41 (18.40)	12.33 (18.09)	12.17 (17.96)	12.10 (18.14)	12.36 (18.96)	12.61 (19.14)
<i>Counties</i>	—	40	40	40	40	40	40	40	40	40	40	40
<i>Treated Counties</i>	—	—	—	—	5	10	29	29	39	39	40	40
Panel C: VS — Demographic Flow, Hospital Ownership, and Patient Health [¶]												
Population	45,228 (59,205)	45,502 (59,564)	45,884 (60,064)	46,435 (60,778)	47,253 (61,282)	47,684 (59,393)	47,863 (58,842)	48,792 (59,597)	49,147 (59,858)	49,571 (60,151)	48,935 (59,129)	49,228 (58,987)
Crude birth rate per 1k pop.	19.38 (2.38)	18.45 (2.01)	18.54 (2.24)	18.66 (1.98)	18.08 (1.86)	17.46 (2.11)	17.42 (2.32)	17.42 (2.19)	16.19 (1.94)	15.69 (2.21)	15.64 (1.44)	14.64 (1.48)
Crude death rate per 1k pop.	7.02 (1.14)	6.97 (1.01)	7.19 (1.01)	7.43 (1.06)	7.28 (1.20)	7.67 (1.31)	7.57 (1.28)	7.44 (1.19)	7.55 (1.15)	7.23 (1.10)	7.16 (1.11)	7.21 (1.00)
Crude net migration rate per 1k pop.	— (—)	-5.51 (2.28)	-3.04 (2.72)	0.66 (2.51)	8.56 (29.54)	10.90 (67.48)	-2.87 (26.57)	13.48 (20.11)	-1.47 (17.10)	1.91 (17.97)	-9.89 (78.74)	1.04 (8.35)
Government hospital birth share (%)	65.8 (11.3)	65.9 (11.2)	65.3 (10.9)	64.3 (10.7)	63.2 (10.7)	59.1 (12.1)	51.6 (15.2)	45.2 (15.3)	36.6 (14.6)	33.6 (18.6)	30.6 (19.8)	32.1 (22.7)
Private hospital birth share (%)	33.8 (11.4)	33.8 (11.3)	34.4 (11.0)	35.5 (10.7)	36.5 (10.9)	39.6 (11.6)	48.1 (15.3)	54.6 (15.3)	63.1 (14.6)	66.2 (18.6)	69.2 (19.9)	67.8 (22.8)
Infant mortality rate per 1k live births	12.74 (6.09)	12.67 (5.30)	12.71 (4.47)	13.40 (5.45)	11.84 (4.97)	14.28 (7.33)	9.91 (4.90)	12.10 (6.42)	10.28 (5.13)	10.79 (4.30)	10.15 (5.45)	9.62 (5.04)
Low birth weight rate (%)	9.28 (1.62)	9.25 (1.79)	9.48 (2.02)	9.64 (1.66)	9.85 (1.43)	10.12 (1.90)	10.49 (1.90)	11.06 (2.32)	10.98 (2.24)	11.36 (2.46)	10.96 (2.48)	11.48 (4.07)
<i>Counties</i>	78	78	78	78	78	78	78	78	78	78	78	78
<i>Treated Counties</i>	—	—	—	—	10	22	52	52	77	77	78	78
Panel D: PR DOL — Macro Labor Market [#]												
Labor force / pop. (%; Figure 3)	30.5 (4.6)	31.4 (4.7)	32.0 (5.1)	32.2 (5.1)	31.4 (4.8)	31.8 (4.9)	32.4 (5.1)	32.5 (5.0)	32.3 (5.2)	31.3 (5.2)	31.5 (4.4)	30.9 (4.3)
Unemployment rate (%; U / LF)	16.9 (4.7)	18.4 (4.9)	19.3 (5.1)	20.1 (5.1)	17.1 (4.6)	16.0 (4.7)	15.8 (4.5)	16.1 (4.8)	15.8 (4.6)	14.2 (4.7)	11.6 (2.2)	13.1 (2.7)
<i>Counties</i>	78	78	78	78	78	78	78	78	78	78	78	78
<i>Treated Counties</i>	—	—	—	—	10	22	52	52	77	77	78	78

Means with standard deviations in parentheses. Sample is the PR balanced-panel counties (78 total); per-dataset counts below). [†] *Panel A (QCEW)*: Overall Healthcare = NAICS 621 (Ambulatory) + 622 (Hospitals) + 623 (Nursing/Residential). All wage variables are CPI-deflated using 1982-1984 as the base year. [‡] *Panel B (CMS)*: Provider counts are point-in-time totals from the CMS Provider of Services file; Government / Private split is the contemporaneous ownership classification. The trajectory rows (Always Government, Always Private, Gov. → Private switchers) classify each provider over its full panel history and are stable across years. CMS balanced panel 1991-2000: 40 counties. *CMS provider mix (provider-years, 1991-2000)*: The CMS county panel summarizes 2,586 provider-year observations from 296 unique facilities. By ownership: 454 (17.6%) government, 872 (33.7%) private for-profit, 1,024 (39.6%) private nonprofit, 236 (9.1%) private-unspecified. By facility category, 36.8% are Hospitals (CMS POS code 01); the remaining provider categories are Hospital (36.8%); Home Health Agency (19.8%); Hospice (18.1%); ESRD Facility (11.9%); Ambulatory Surgical Center (7.8%); Other (5.6%). [¶] *Panel C (VS)*: Crude birth rate is live births per 1,000 mid-year population. Crude death rate is deaths per 1,000 mid-year population. Crude net migration rate is the residual of the demographic-balancing equation: $[(\text{Pop}_t - \text{Pop}_{t-1}) - (\text{Births}_t - \text{Deaths}_t)] / \text{Pop}_{t-1} \times 1000$, NA in 1990 (no $t - 1$ population). Government and private hospital birth shares are percentages of all live births. Low birth weight rate is the percentage of live births under 2,500g. VS balanced panel 1990-2001: 78 counties. [2pt][#] *Panel D (PR DOL)*: Labor force / pop. is labor force divided by total county population $\times 100$; employment / pop. is employment divided by total county population $\times 100$. Both use the non-standard denominator (total population) rather than the BLS-standard civilian non-institutional population aged 16+ (CNIWAP), which is not available at the county level for the 1990-2001 panel. The unemployment rate is reported as BLS-standard (unemployed / labor force $\times 100$). PR DOL balanced panel 1990-2001: 78 counties. Across all 936 county-year observations: BLS-style U/LF mean = 16.19%; alternative U/Pop mean = 5.04%; difference = 11.16 pp; Pearson correlation between the two series = 0.878. The two series differ in level but not in trends. *Pre-reform (1990-1993) cross-county medians used as thresholds in heterogeneity figures (Main Figure 5; Appendix Figures A.5-A.7; Online Appendix Figures OA.21-OA.22)*: Government hospital birth share = 63.2%; Private hospital birth share = 36.3%; Healthcare employment per 1k pop. = 1.51; Population = 30,362; Unemployment rate = 18.6%. Each median is computed across counties after first averaging the outcome over the 1990-1993 window within each county.

Table 2: Pre-Reform Balance Across Treatment Cohorts (1990–1993 Means)

Variable	Cohort 1 (1994) <i>Mean</i>	Cohort 2 (1995) <i>Mean</i>	Cohort 3 (1996) <i>Mean</i>	Cohort 4 (1998) <i>Mean</i>	Cohort 5 (2000) <i>Mean</i>	<i>F</i> -test <i>p-value</i>	<i>N</i>
Total population	43,889	34,467	37,871	45,489	443,627	<0.001	78
Log population	10.124	10.287	10.330	10.427	13.003	0.013	78
Share urban (top-20 by 1990 density)	0.600	0.417	0.433	0.560	1.000	0.638	78
Gov. hospital birth share (%)	73.57	66.70	67.17	59.30	61.64	0.003	78
HC employment per 1k pop.	4.01	2.28	2.64	2.87	29.72	<0.001	60
Labor force participation rate (%)	34.96	29.79	28.96	33.90	35.47	<0.001	78
Unemployment rate (%)	15.23	19.60	20.97	17.12	10.70	<0.001	78
Crude net migration rate (per 1k)	-3.80	-2.87	-3.25	-1.48	1.89	0.002	78
Infant mortality rate (per 1k)	13.59	11.21	13.74	12.10	13.02	0.139	78
Crude birth rate (per 1k)	19.90	19.11	19.39	17.43	17.22	<0.001	78
Agriculture, forestry & fishing emp. per 1k	1.451	16.195	8.422	15.991	–	0.393	25
Transportation & warehousing emp. per 1k	3.595	0.623	0.644	1.287	9.811	0.037	78
Arts, entertainment & recreation emp. per 1k	4.745	–	0.281	0.342	2.145	0.525	11
<i>Counties</i>	10	12	30	25	1		78

Cell entries are 1990–1993 means of pre-reform covariates by treatment cohort. Cohorts 1–5 correspond to reform-rollout years 1994, 1995, 1996, 1998, and 2000 respectively (no rollouts in 1997 or 1999). The *F*-test column reports the *p*-value of the one-way ANOVA test for the joint null hypothesis that all five cohort means are equal (`aov(var ~ factor(cohort))`); a small *p*-value indicates that the listed covariate differs systematically across cohorts and is therefore not balanced under the rollout schedule. **Sources.** Population, gov. hospital birth share, infant mortality, crude birth rate, and crude net migration rate are 1990–1993 means from the Puerto Rico Department of Health vital statistics. Healthcare employment per 1,000 residents (NAICS Overall Healthcare = 621 + 622 + 623) and the three placebo-industry rows (Agriculture/Forestry/Fishing = NAICS 11; Transportation & Warehousing = NAICS 48-49; Arts/Entertainment/Recreation = NAICS 71) are 1990–1993 means computed from the QCEW (BLS) county-quarter panel: aggregated across ownership classes, averaged to annual frequency requiring at least 3 reporting quarters per year, then divided by the population. Labor force participation and unemployment rates are 1990–1993 means from the PR Department of Labor. The urban indicator equals one for the 20 counties with the highest 1990 population density and zero otherwise; the cohort-cell entry is the share of urban counties within the cohort. The three placebo-industry rows (NAICS 11, 48-49, 71) are the same industries used in the QCEW placebo robustness analysis (Online Appendix Section 7)

Table 3: Effects of the Reform on Hospital Ownership Composition (First Stage)

	(1) Gov. Hospital Birth Share (%)	(2) Priv. Hospital Birth Share (%)
ATT	-14.79*** (1.94)	15.19*** (1.93)
Pre-treatment mean of dep. var.	62.55	37.04
ATT as % of pre-treatment mean	-23.6%	41.0%
<i>N</i> counties	78	78

Estimates use the Callaway and Sant’Anna (2021) ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. The pre-treatment mean of the dependent variable is computed across treated counties using observations from the cohort-specific pre-treatment period (event time < 0). Government and private hospital birth shares are defined as the count of live births occurring at facilities of each ownership class divided by the total count of live births in the county-year, expressed in percent. The two shares are not exact complements because a small tail of births (typically $< 1\%$) is recorded as non-hospital or with unspecified facility ownership.

Table 4: Reform Effects on Healthcare Employment, Wages, and Establishments

	Healthcare (NAICS 621+622+623)			NAICS 621 (Ambulatory)		
	(1) Empl. per 1k	(2) Wages/Empl. per 1k	(3) Estabs. per 1k	(4) Empl. per 1k	(5) Wages/Empl. per 1k	(6) Estabs. per 1k
ATT	-0.511*** (0.157)	0.630 (3.734)	-0.050*** (0.019)	-0.502*** (0.125)	0.867 (3.682)	-0.034* (0.018)
Pre-treatment mean of dep. var.	3.923	38.962	0.759	2.895	38.889	0.747
ATT as % of pre-treatment mean	-13.0%	1.6%	-6.6%	-17.3%	2.2%	-4.6%
<i>N</i> counties	59	59	59	59	59	59

Estimates use the Callaway and Sant’Anna (2021) ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. The pre-treatment mean of the dependent variable is computed across treated counties using observations from the cohort-specific pre-treatment period (event time < 0). The reference period for event-time aggregation is $t = -1$ (year before treatment).

Table 5: Effects of The Reform on Patient Health (Short-Horizon) and Hospital Capacity

	Patient Health (VS)		Hospital Capacity (CMS)	
	(1) IMR (per 1k live births)	(2) Low Birth Weight (%)	(3) Beds per 1k pop.	(4) Services per 1k pop.
ATT	-1.205 (0.928)	0.060 (0.230)	-0.265*** (0.065)	-0.024** (0.010)
Pre-treatment mean of dep. var.	12.620	9.717	3.389	0.487
ATT as % of pre-treatment mean	-9.5%	0.6%	-7.8%	-4.9%
<i>N</i> counties	78	78	40	40

Estimates use the Callaway and Sant'Anna (2021) ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. The pre-treatment mean of the dependent variable is computed across treated counties using observations from the cohort-specific pre-treatment period (event time < 0). The reference period for event-time aggregation is $t = -1$ (year before treatment).

Table 6: Worker-Side Adjustment Margins to The Reform

	(1) Labor force / pop. (%)	(2) Crude Net Migration (per 1k)	(3) Crude Birth Rate (per 1k)
ATT	-1.110*** (0.241)	-14.651** (7.389)	-0.564** (0.225)
Pre-treatment mean of dep. var.	31.895	1.059	18.267
ATT as % of pre-treatment mean	-3.5%	-1382.9%	-3.1%
<i>N</i> counties	78	78	78

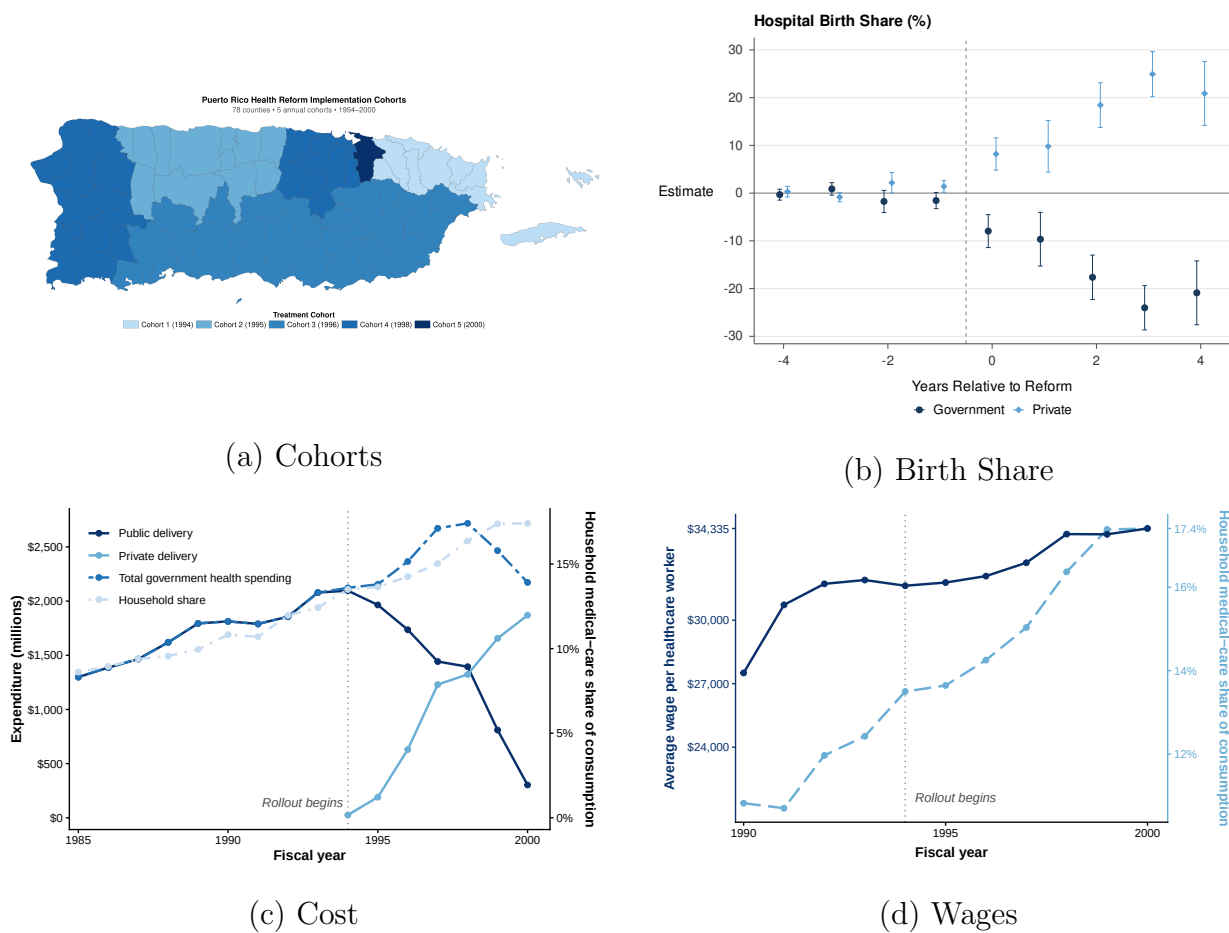
Estimates use the Callaway and Sant'Anna (2021) ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. The pre-treatment mean of the dependent variable is computed across treated counties using observations from the cohort-specific pre-treatment period (event time < 0). The reference period for event-time aggregation is $t = -1$ (year before treatment).

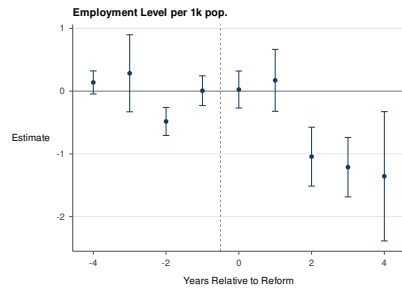
Table 7: Heterogeneity in the Effects by Pre-Reform Public-Hospital Exposure

	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
<i>Below-median pre-reform government hospital birth share</i>				
ATT	-15.49*** (2.44)	-0.346 (0.233)	-9.206 (10.814)	-1.454 (1.101)
Pre-treatment mean	55.49	4.911	2.772	11.577
ATT as % of pre-mean	-27.9%	-7.0%	-332.1%	-12.6%
<i>N</i> counties	39	36	39	39
<i>Above-median pre-reform government hospital birth share</i>				
ATT	2.41 (9.15)	0.145 (0.268)	-23.891** (12.047)	0.660 (1.329)
Pre-treatment mean	71.21	2.069	-1.126	13.925
ATT as % of pre-mean	3.4%	7.0%	2121.2%	4.7%
<i>N</i> counties	39	23	39	39

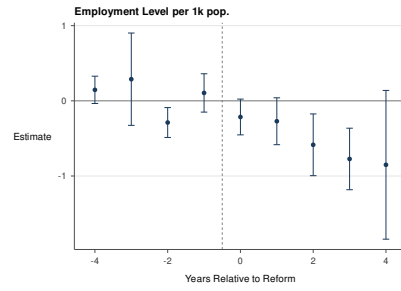
Notes: Estimates use the Callaway and Sant'Anna (2021) doubly-robust ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. Counties are partitioned at the cross-county median of the 1990–1993 pre-reform government hospital birth share (median = 63.24%). *Below-median* (39 counties) = lower pre-reform reliance on the public hospital system; *Above-median* (39 counties) = higher pre-reform reliance.

10 Figures

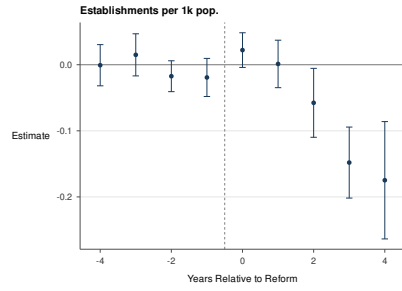




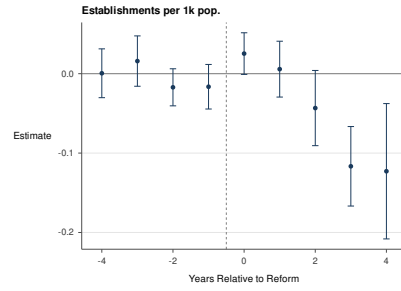
(a) Employment (per 1k) - 621+622+623



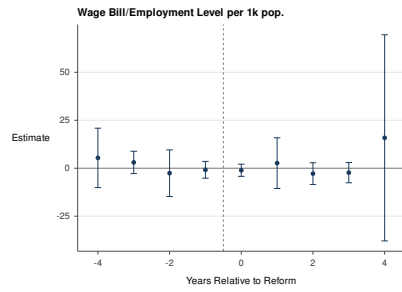
(b) Employment (per 1k) - 621



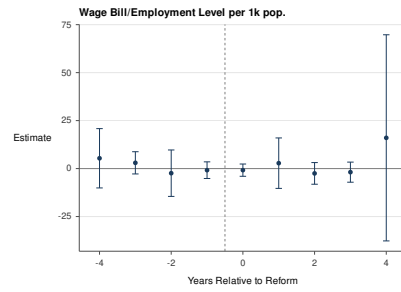
(c) Establishments (per 1k) - 621+622+623



(d) Establishments (per 1k) - 621

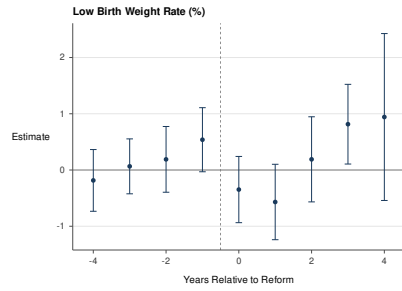


(e) Wage Bill/Employee (per 1k) - 621+622+623

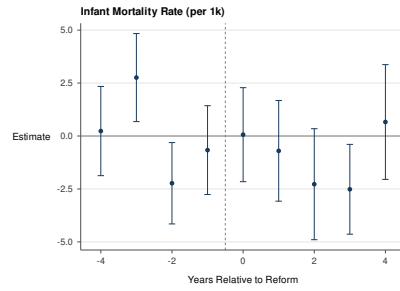


(f) Wage Bill per Employee (per 1k) - 621

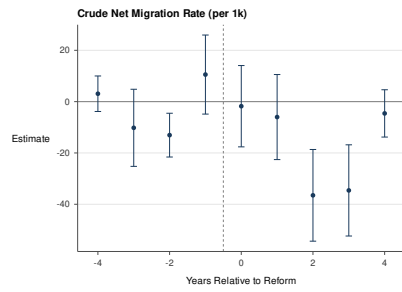
Figure 2: Healthcare Labor-Market Contraction



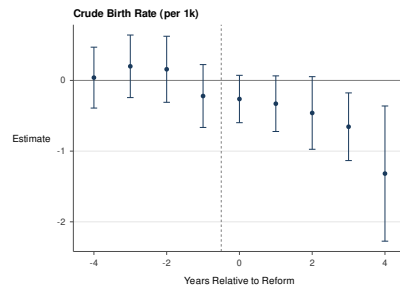
(a) Low Birth Weight



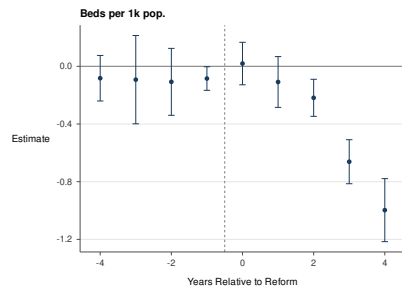
(b) Infant Mortality Rate



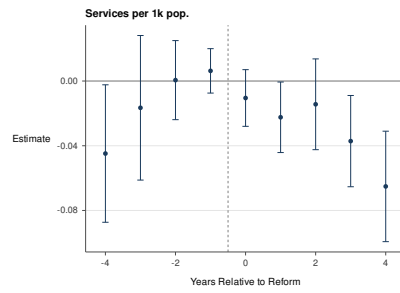
(c) Net Migration Rate



(d) Birth Rate

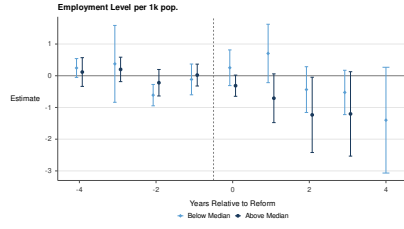


(e) Beds

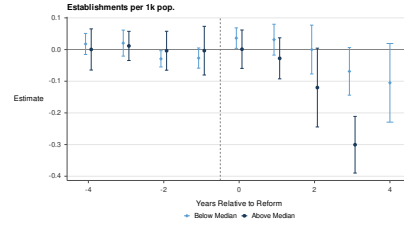


(f) Services

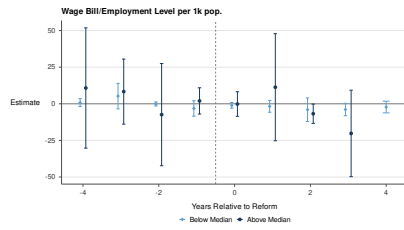
Figure 3: Patient Health



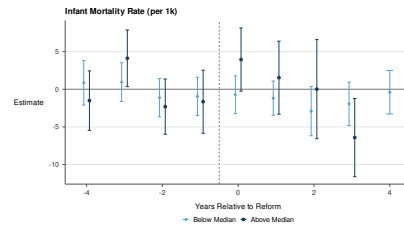
(a) Employment (per 1k) - 621+622+623



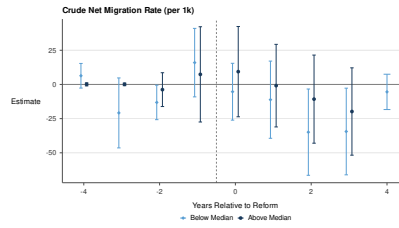
(b) Establishments (per 1k) - 621+622+623



(c) Wage Bill per Employee (per 1k) - 621+622+623

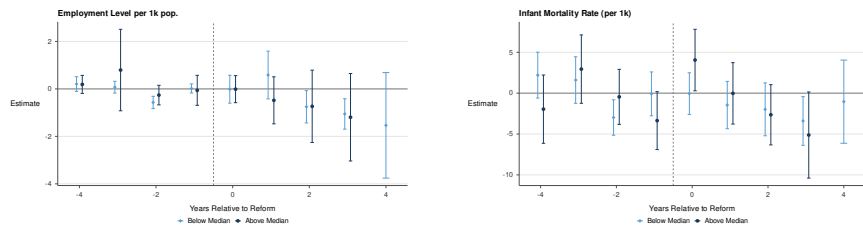


(d) Infant Mortality Rate (per 1k)

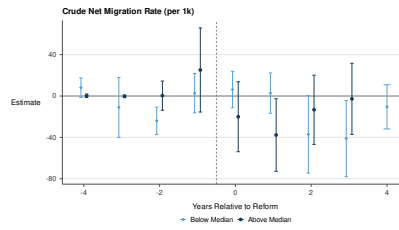


(d) Net Migration Rate

Figure 4: Heterogenous Effects by Poverty Rate



(a) Employment (per 1k) - 621+622+623 (b) Infant Mortality Rate (per 1k)



(c) Net Migration Rate (per 1k)

Figure 5: Heterogeneous Effects by Pre-Reform Gov. Hospital Exposure

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Appendix

The Appendix collects the supplementary tables referenced in the body. Each table is referenced from a footnote to the relevant subsection of the main text. The Online Appendix (Section 10) hosts the remaining robustness, sample-restriction, and subgroup specifications.

A.1 Estimator-Class Robustness

This subsection reports a four-estimator triangulation on the six primary outcomes (TWFE; the ETWFE estimator of Wooldridge, 2025; the doubly robust estimator of Callaway & Sant’Anna, 2021, C&S; and Borusyak et al., 2024, BJS). Pre-treatment parallel trends hold for the labor-market outcomes under all four estimators. The ETWFE, C&S, and BJS estimators agree on the sign and magnitude of the first-stage hospital-ownership share, healthcare employment, labor-force participation, and the hospital-bed contraction. TWFE reverses sign on the outcomes with the most treatment-effect heterogeneity (healthcare employment, infant mortality, and the low-birth-weight share). The crude net migration rate is the only outcome on which the four estimators disagree on sign.

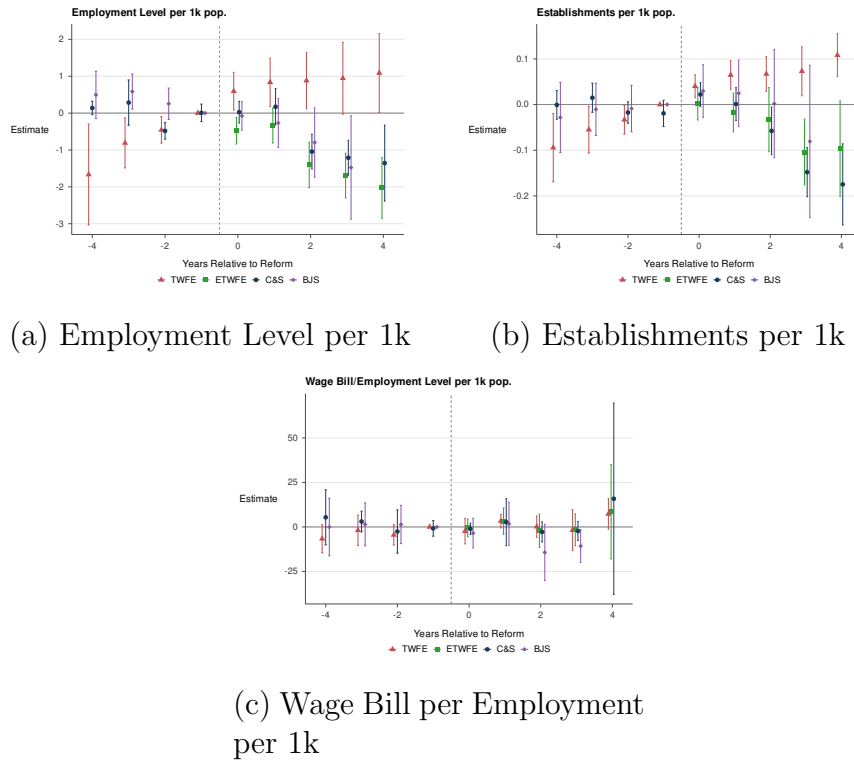


Figure A.1: Overall Healthcare Labor Market - Estimator Comparison

Table A.1: Four-Estimator Robustness: ATT Estimates Under TWFE, ETWFE, C&S, and BJS

	(1) TWFE	(2) ETWFE	(3) C&S	(4) BJS
<i>Gov. Hospital Birth Share (%)</i>				
ATT	-8.08*** (1.89)	-16.52*** (1.60)	-14.79*** (1.94)	-11.25*** (1.69)
ATT as % of pre-mean	-12.9%	-26.4%	-23.6%	-18.0%
Pre-treatment mean counties			62.55 78	
<i>HC Employment per 1k pop. (NAICS 621+622+623)</i>				
ATT	0.868** (0.352)	-1.008*** (0.181)	-0.511*** (0.157)	-0.655*** (0.217)
ATT as % of pre-mean	22.1%	-25.7%	-13.0%	-16.7%
Pre-treatment mean counties			3.923 59	
<i>Labor force / pop. (%)</i>				
ATT	0.059 (0.350)	-1.155*** (0.231)	-1.110*** (0.241)	-0.712*** (0.235)
ATT as % of pre-mean	0.2%	-3.6%	-3.5%	-2.2%
Pre-treatment mean counties			31.895 78	
<i>Crude Net Migration Rate (per 1k pop.)</i>				
ATT	-5.689 (15.281)	-1.525 (2.931)	-14.651** (7.389)	4.513 (3.523)
ATT as % of pre-mean	-537.1%	-144.0%	-1382.9%	426.0%
Pre-treatment mean counties			1.059 78	
<i>Infant Mortality Rate (per 1k live births)</i>				
ATT	1.973** (0.815)	-1.524** (0.599)	-1.205 (0.928)	-0.513 (0.727)
ATT as % of pre-mean	15.6%	-12.1%	-9.5%	-4.1%
Pre-treatment mean counties			12.620 78	
<i>Beds per 1k pop.</i>				
ATT	-0.387 (0.333)	-0.443*** (0.072)	-0.265*** (0.065)	-0.189*** (0.056)
ATT as % of pre-mean	-11.4%	-13.1%	-7.8%	-5.6%
Pre-treatment mean counties			3.389 40	

Notes: Four estimators are reported. TWFE: two-way fixed-effects regression with county and year FEs and event-time dummies. ETWFE: Wooldridge (2021) Extended Two-Way Fixed Effects. C&S: Callaway and Sant'Anna (2021). BJS: Borusyak, Jaravel & Spiess (2024). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors.

A.2 Excluding San Juan

Reports the main-vs.-excluded-San-Juan specifications for the QCEW labor-market and CMS hospital-capacity outcomes. The healthcare-employment and bed-capacity estimates attenuate substantially when San Juan is removed; the first-stage ownership-transition estimate (Online Appendix Table OA.3) survives the exclusion at conventional significance.

Table A.2: Sample-Specification Robustness: Excluding San Juan and the Unbalanced-Panel Variant

	HC Empl. per 1k pop.			Beds per 1k pop.			Infant Mortality Rate		
	(1) Bal.	(2) Excl. SJ	(3) Unbal.	(4) Bal.	(5) Excl. SJ	(6) Unbal.	(7) Bal.	(8) Excl. SJ	(9) Unbal.
ATT	-0.511*** (0.157)	-0.168 (0.191)	-0.511*** (0.157)	-0.265*** (0.065)	0.090 (0.140)	-0.265*** (0.065)	-1.205 (0.928)	0.781 (1.237)	-1.205 (0.928)
Pre-treatment mean of dep. var.	3.923	3.161	3.892	3.389	3.033	2.980	12.620	12.624	12.620
ATT as % of pre-treatment mean	-13.0%	-5.3%	-13.1%	-7.8%	3.0%	-8.9%	-9.5%	6.2%	-9.5%
counties	59	58	60	40	39	54	78	77	78

Notes: Estimates use the Callaway and Sant'Anna (2021). Significance levels: * $p < 0.10$, ** $p < 0.05$,

*** $p < 0.01$, based on two-sided tests with cluster-robust standard errors.

A.3 Section 936 Manufacturing Control

Reports the main healthcare-employment estimate with and without conditioning on pre-reform manufacturing exposure (1990–1993 mean of NAICS 31–33 employment per thousand at the county level). The C&S no-control estimate on the 54-county sample is -14.7 percent of pre-mean ($p < 0.01$); the C&S estimate with the strictly-pre-1996 manufacturing-exposure covariate is -6.8 percent of pre-mean (statistically insignificant). The TWFE columns are reported for direct horse-race comparison and return the positive point estimates expected under Goodman-Bacon contamination on heterogeneous-effect outcomes. The main contraction survives conditioning on §936 exposure at attenuated magnitude.

Table A.3: IRS Section 936 Manufacturing-Control Robustness: Healthcare Employment per 1,000 Residents under TWFE and Callaway-Sant’Anna, with and without Manufacturing Exposure as a Control

	TWFE		Callaway-Sant’Anna	
	(1) No control	(2) With manuf. control	(3) No control	(4) With baseline manuf. exposure
ATT on HC empl. per 1k	0.173 (0.198)	0.182 (0.215)	-0.484*** (0.164)	-0.224 (0.233)
ATT as % of pre-mean	5.3%	5.5%	-14.7%	-6.8%
Coef. on contemp. manuf. per 1k	— —	-0.003 (0.013)	— —	— —
counties	54	54	54	54

Notes: Cluster-robust standard errors at the county level. Outcome: healthcare employment per 1,000 residents (NAICS 621+622+623). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. **Estimators.** Columns 1–2 report TWFE event-study estimates with county and year fixed effects.

A.4 Welfare Bounds

Translates the main supply contraction and the patient-health response into comparable units in constant 2025 USD, with parenthetical translations into the 1982–1984 USD price base of the calculation. The cells span two parameter dimensions: the healthcare-employment-contraction estimate (main C&S versus §936-conditional C&S, the latter from Table A.3) and whether the patient-health response is valued at zero (the full-panel IMR null) or at the high-substitution-half magnitude. Because the full-panel infant-mortality effect is insignificant, the health-valued cells enter only as an upper bound and are evaluated across the four high-substitution partitions of Section 6.4, whose above-median IMR estimates range from -2.03 to -3.12 deaths per thousand live births. The zero-health cells, which rest entirely on the avoided supply cost, are \$967 million under the headline supply estimate (\$69 per resident per year; \$300.8 million in 1982–1984 dollars) and \$844 million under the §936-conditional estimate (\$60 per resident per year; \$262.5 million in 1982–1984 dollars). The health-valued cells add the monetized mortality reduction and, across the four partitions, range from \$3.4 to \$4.7 billion in 2025 dollars (\$1.05 to \$1.46 billion in 1982–1984 dollars); the welfare table reports the mean-partition value, about \$4.0 billion in 2025 dollars (\$1.26 billion in 1982–1984 dollars), as the headline upper bound. The exercise is partial-equilibrium and back-of-envelope, not a structural welfare calculation.

Table A.4: Welfare Bounds: Cost Savings, Patient-Health Benefits, and Net Welfare

	Lower bound ($\Delta\text{IMR} = 0$) (1982-84 \$M)	Upper bound (ΔIMR high-substitution half) (1982-84 \$M)
<i>Panel A: Headline ATT (Tables 2 and 4)</i>		
Cost saving (\$M)	300.8	300.8
Health benefit (\$M)	0.0	958.7
Net welfare (\$M)	300.8	1,259.5
Per capita per year (\$)	21.5	90.0
<i>Panel B: § 936-Conditional ATT (Table A.4 col. 4)</i>		
Cost saving (\$M)	262.5	262.5
Health benefit (\$M)	0.0	958.7
Net welfare (\$M)	262.5	1,221.2
Per capita per year (\$)	18.7	87.2

Notes: Partial-equilibrium back-of-envelope welfare bound; **not** a structural calculation. Body cells are in real 1982-84 USD (CPI-U all-items, 1982-84 = 100); the headline upper bound corresponds to approximately \$4,048 M in 2025 USD. **Cost-saving inputs.** $\Delta\text{beds}/1\text{k} = -0.265$ (Table 4); $\Delta\text{HC empl.}/1\text{k} = -0.511$ headline (Table 2) and -0.224 § 936-conditional (Table A.4 col. 4). Cost per bed-year (\$62,756, 1982-84 USD) is the implied 1990–1993 mean hospital wage bill from CMS POS and QCEW NAICS 621+622+623, scaled by 2.1 (U.S. HCRIS hospital total-cost-to-wage-bill ratio). **Patient-health benefit.** VSL = \$10 M (U.S. EPA standard, 2024 USD), rebased to 1982-84 USD as $\approx \$3.19$ M. $\Delta\text{IMR} = -2.59$ per 1,000 live births in the high-substitution half (mean across pre-reform population density, total population, HC employment intensity, and commuting inflow; Tables A.7, OA.9–OA.11); a 0.5 multiplier rescales island-wide births to the high-substitution half.

A.5 Spillover-Excluded Sample (QCEW)

Reports the QCEW labor-market specifications on the non-border subsample ($n = 27$ counties retained out of 78). The healthcare-employment, wage, and establishment estimates are statistically indistinguishable from zero on this restricted sample; the first-stage hospital-ownership-share estimate (Online Appendix Table OA.5) survives the restriction at -12.14 percentage points ($p < 0.01$, -19.4 percent of pre-mean).

Table A.5: Spillover-Excluded Sample: QCEW Healthcare Headline Outcomes

	(1) Empl. per 1k	(2) Wages/Empl. per 1k	(3) Estabs. per 1k
ATT	-0.054 (0.318)	3.083 (9.307)	0.021 (0.037)
Pre-treatment mean of dep. var.	3.193	42.066	0.787
ATT as % of pre-treatment mean	-1.7%	7.3%	2.7%
counties	27	27	27

Notes: Callaway and Sant’Anna (2021) doubly-robust ATT with not-yet-treated controls; reference period $t = -1$; cluster-robust standard errors at the county level via multiplier bootstrap (1,000 reps). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. **Spillover-exclusion sample.** Restricted to the 27 counties that share no boundary with a different-cohort county.

A.6 Heterogeneity by Pre-Reform Population Density

Partitions the seventy-eight counties at the median pre-reform population density. The infant-mortality estimate in the above-median (urban) cell is -2.027 per thousand live births ($p < 0.05$, -16.9 percent of pre-mean). The first of four heterogeneity dimensions on which the IMR signal is stronger than the baseline-exposure cut reported in Table 7.

Table A.6: Heterogeneity by Pre-Reform Population Density (Urban vs. Rural Median Split)

	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
<i>Below-median pre-reform population density (urban vs. rural)</i>				
ATT	-1.27 (5.27)	-0.105 (0.262)	-7.565 (10.268)	2.436 (1.956)
Pre-treatment mean	67.13	2.074	0.587	13.273
ATT as % of pre-mean	-1.9%	-5.1%	-1288.3%	18.4%
counties	39	24	39	39
<i>Above-median pre-reform population density (urban vs. rural)</i>				
ATT	-18.21*** (2.36)	-0.418* (0.219)	-12.513 (11.042)	-2.027** (0.892)
Pre-treatment mean	58.11	5.172	1.514	11.993
ATT as % of pre-mean	-31.3%	-8.1%	-826.8%	-16.9%
counties	39	35	39	39

Notes: Estimates use the Callaway and Sant’Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. The pre-treatment mean of the dependent variable is computed across treated counties using observations from the cohort-specific pre-treatment period (event time < 0). Pre-reform population density = (1990–1993 mean municipio population) / (land area in km²). Counties partitioned at the cross-municipio median; below-median (‘Rural’) = 24 counties, above-median (‘Urban’) = 35 counties.

A.7 Per-Capita versus Raw-Level Specifications

Reports per-thousand-resident and $\log(1+y)$ raw-level specifications side-by-side for QCEW healthcare employment, establishments, and the wage bill. Healthcare employment per thousand falls by 13.0 percent ($p < 0.01$); $\log(1+y)$ raw-level employment grows by approximately 19.0 percent ($p < 0.05$). The per-capita decline reflects population growth that outpaced the growth of the local healthcare workforce; the per-capita normalization is the welfare-relevant scaling adopted throughout (Section 4). The distributional finding of the paper rests on the per-capita denominator and is not a feature of the absolute headcount.

Table A.7: QCEW Healthcare Headline Outcomes: Per-1,000 vs. Log-Level

	HC Employment		HC Establishments		HC Wage Bill	
	Per 1,000	Log(1+x)	Per 1,000	Log(1+x)	Per 1,000	Log(1+x)
ATT	-0.511*** (0.157)	0.190** (0.074)	-0.050*** (0.019)	0.144*** (0.048)	0.630 (3.734)	0.247 (0.263)
Pre-treatment mean	3.923	4.293	0.759	3.161	38.962	11.000
Implied % effect	-13.0%	19.02%	-6.6%	14.38%	1.6%	24.69%

Notes: Estimates use the Callaway and Sant'Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors.

A.8 Labor Force

The labor-force participation rate contracts over the rollout period, consistent with the negative labor-force-participation estimate in Section 6.3. Because the unemployment rate does not move over the same window, the adjustment operates on participation rather than on measured unemployment.

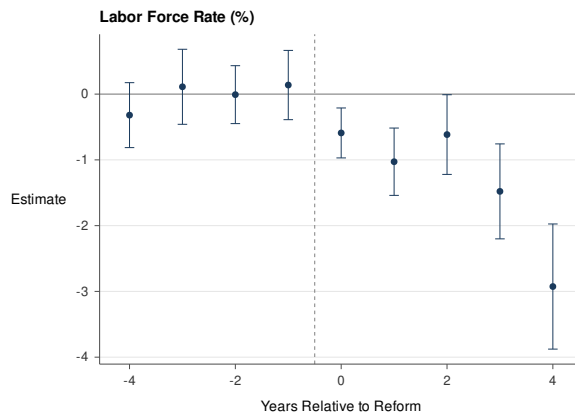


Figure A.2: Labor Force

A.9 Financing Detail

The 1993 reform replaced the Department of Health line-item appropriation that had funded the pre-reform delivery system with a multi-source aggregate funding pool flowing through ASES to the contracting insurers. Realized program financing for fiscal year 2002–2003 totaled approximately \$2.24 billion in 2025 dollars, drawn in approximate proportions of seventy-three percent from the Commonwealth General Fund, fifteen percent from federal Medicaid, eleven percent from the counties, and two percent from the State Children’s Health Insurance Program (CHIP) (Pan American Health Organization, 2007). An independent reconstruction from the Office of Management and Budget agency operating-budget volumes places the ASES program at \$2.27 billion in fiscal year 2002 and \$2.18 billion in fiscal year 2003 (2025 dollars), bracketing the \$2.24 billion administrative-financing figure within a few percent from a separate source (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 2004, 2005).³¹ About seventy to seventy-five percent of the ASES outlay is premium pass-through to the contracting managed-care organizations, consistent with the program operating as a financing pool rather than a direct provider (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 2007). The federal Medicaid share is constrained by the Section 1108(g) territorial Medicaid cap and the fixed Federal Medical Assistance Percentage (FMAP) applicable to Puerto Rico, both of which produce a federal contribution per enrollee substantially below what Puerto Rico would receive under the income-conditioned FMAP formula applied to the fifty states (Kaiser Family Foundation, 2024; Medicaid and CHIP Payment and Access Commission, 2020).

The composition of that financing shifted as the public delivery agencies were dismantled. Across fiscal years 1996 to 1999 the Commonwealth General Fund supporting the health delivery and regulatory sector, excluding ASES, fell from about \$985 million to \$572 million while own-source revenue rose from about \$382 million to \$483 million and the federal contribution held near \$650 million; budgeted positions in the same sector fell from 26,905 to 18,062, a 33 percent reduction over the period (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1998).³²

The program continues to operate under the names *Mi Salud* and *Vital*, covering approximately 1.3 million Medicaid and CHIP enrollees as of 2026 (Centers for Medicare and Medicaid Services, 2026).

³¹The two series rest on different bases, an all-funds operating-budget appropriation in the budget volumes and realized program financing in Pan American Health Organization (2007), so their agreement to within a few percent is a cross-source check rather than an identity. In nominal terms these are \$1.271 billion (fiscal year 2002) and \$1.248 billion (fiscal year 2003) from the budget volumes and \$1.282 billion (fiscal years 2002–2003) from Pan American Health Organization (2007); the text deflates them to 2025 dollars with the U.S. CPI-U.

³²This is the public-sector budgeted headcount, distinct from the QCEW total-employment-per-thousand outcome estimated in Section 6; it parallels that contraction on the public-employment side and is corroborative rather than the same series. The 1998 figure in this grouping is a current-budget (*vigente*) value, so the endpoints rather than the intervening years carry the comparison. In nominal terms the General Fund fell from \$480 million to \$296 million, own-source revenue rose from \$186 million to \$250 million, and the federal contribution held near \$330 million (fiscal years 1996 and 1999); the text deflates these to 2025 dollars with the U.S. CPI-U. In fiscal year 1999 the General Fund supporting this health sector (\$296 million) was about 4 percent of the Commonwealth’s \$6.7 billion General Fund (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1999).

The household side of medical spending followed its own long-run path over this period: the medical-care share of personal consumption continued the secular rise it had begun well before the reform, from about 5.6 percent in 1971 to 12.4 percent in 1993 and 17.4 percent by fiscal year 2000, even as spending on the public delivery agencies fell (Figure A.4). We read this as descriptive context for the financing reallocation, not as evidence that the reform shifted costs onto households.

A.10 Data Construction

This subsection reports the construction details for each of the six data sources summarized in Section 4.

QCEW (1990–2000). The Quarterly Census of Employment and Wages provides county-quarter establishment cells with average monthly employment, total quarterly wages, and active establishment counts by NAICS code and ownership class (federal, state, and local government, and private). The annualization procedure proceeds in two steps. Within each county-quarter cell, employment, wages, and establishments are summed across ownership classes to produce a quarter-level total. Across quarters within a county-year, employment and establishments are averaged and wages are summed; cells with fewer than three reporting quarters are dropped on the grounds that they reflect missing rather than zero health-care activity. Per-thousand-resident outcomes are constructed by dividing each headcount by the contemporaneous Vital Statistics population in thousands. The wage-per-employee intensive-margin outcome is the annualized real wage bill divided by annualized employment, expressed in 2024 dollars using BLS CPI-U annual-average deflators. The balanced-cohort QCEW healthcare panel covers 59 counties and 649 county-year observations.

CMS Provider of Services (1991–2000). The CMS POS public-use files provide annual provider-level records of Medicare-certified facilities with full-time-equivalent counts, certified beds, services rendered, and ownership classification (government, voluntary nonprofit, proprietary). Provider-level records are aggregated to the county-year level by summing across providers operating in each county. Of 296 unique providers tracked across the 1991–2000 panel, 32 are continuously government-owned, 225 are continuously private, and 12 switched ownership within the panel window; the small switcher count underscores the partial coverage of the privatization shock at the Medicare-certified layer. The balanced CMS Track A panel covers 40 counties and 400 county-year observations.

Vital Statistics (1990–2000). The Puerto Rico Department of Health Vital Statistics annual records provide county-year totals of live births, deaths, infant deaths, and end-of-year population. These records support construction of (i) the crude birth rate, (ii) the infant mortality rate, (iii) the low-birth-weight share, (iv) the government and private hospital birth shares, (v) the crude net migration rate (CNMR), constructed as the demographic-balancing residual

$$\text{CNMR}_{i,t} = \frac{(\text{Pop}_{i,t} - \text{Pop}_{i,t-1}) - (\text{Births}_{i,t} - \text{Deaths}_{i,t})}{\text{Pop}_{i,t-1}} \times 1,000, \quad (4)$$

and (vi) total county population in levels and in logarithms, used as a complementary demographic outcome to (4) that is not constructed as a residual and is therefore less sensitive to compounding measurement error. CNMR estimates are reported in level units (per thousand residents per year) rather than as a percentage of the pre-treatment mean because the CNMR pre-treatment mean is close to zero (+1.06 per thousand). The CNMR is reported alongside total population because the two outcomes can in principle disagree on the sign of the local demographic response if the constituents (births, deaths, populations) are measured with component-wise error; the joint reading is informative about which margin (residual versus level) is the more reliable identification anchor in any given specification.

PR Department of Labor (1990–2000). The Puerto Rico Department of Labor and Human Resources monthly county-level labor-force survey (*Encuesta de Vivienda*) provides annualized county-year measures of the labor force, employment, and unemployment. The unemployment rate is constructed in BLS-standard form (unemployed over labor force). The labor-force-participation rate and the employment-to-population rate are constructed against total county population, since the BLS-standard 16-and-over civilian non-institutional denominator is not available at the Puerto Rican county level for the 1990–2000 panel. The maintained assumption is that the ratio of working-age civilian non-institutional population to total population is not differentially affected by treatment within the analysis window; the partial effect identified on the total-population-denominator rate is then proportional to the partial effect that would be identified on the BLS-standard rate.

NVSS Natality (conception years 1993–2000). The National Center for Health Statistics National Vital Statistics System natality public-use files provide birth-level records with infant birth weight, gestational age at delivery, the count of prenatal visits, the trimester of prenatal-care initiation, the one- and five-minute Apgar scores, and maternal demographic and educational characteristics. Puerto Rico appears in the separate U.S. territory natality file, which begins with the 1994 birth cohort. The empirical specification assigns treatment by conception year, treating a birth if its conception year post-dates the treatment date in the county of maternal residence, rather than by delivery year, on the grounds that the relevant exposure of the prenatal-care production function is the conception-to-delivery window; indexing the 1994-onward birth cohorts this way yields the 1993–2000 conception window used here.

BRFSS (1996–2000). The Centers for Disease Control and Prevention Behavioral Risk Factor Surveillance System provides individual-level adult-survey microdata with self-reported insurance status, self-reported general health (a five-point ordinal scale), an indicator for whether the respondent had been unable to see a doctor in the past twelve months because of cost, and standard demographic covariates. The BRFSS Puerto Rico coverage begins in 1996, after Cohorts 1 and 2 had completed implementation; the BRFSS analysis is therefore restricted to Cohorts 3–5 (1996, 1998, and 2000), for which a pre-treatment observation window exists within the panel.

A.11 Framework Derivation

This subsection reports the formal derivations behind the three-component decomposition (1) stated in Section 3.

Ownership component. Following Hart et al. (1997), the noncontractible margin includes provider effort, time per patient, and continuity of care; the contractible margin is the per-enrollee budget that ASES pays each managed-care organization. Let $\theta \in [0, 1]$ index, in reduced form, the relative weight of quality ($\theta = 1$) versus cost ($\theta = 0$) in the provider’s objective. The parameter summarizes the residual-control result of Hart et al. (1997): the private provider captures the returns to cost reduction and cannot be penalized for cutting quality no contract specifies, so its cost-reduction incentive is stronger and its effective weight on quality lower, $\theta_{\text{public}} > \theta_{\text{private}}$. Letting $L^*(\theta)$ denote optimal labor input, the model implies $L^*(\theta) > 0$, so the ownership-component response on healthcare employment per capita satisfies

$$\Delta L^{\text{ownership}} = L^*(\theta_{\text{private}}) - L^*(\theta_{\text{public}}) \leq 0. \quad (5)$$

The same primitive admits the quality-deterioration corollary referenced in Section 3: non-contractible quality q satisfies $\Delta q \leq 0$ unless the public provider was itself underinvesting on the quality margin relative to the private optimum.

Capitation component. The pre-reform compensation arrangement paid providers fixed government salaries; the post-reform arrangement pays managed-care organizations prospective per-member-per-month capitation, with insurers in turn contracting bilaterally with provider networks. Let $C(L)$ denote the provider’s expected cost as a function of network labor input. Under salary, $C'(L)$ is borne by the principal (the Department of Health); under capitation, it is borne by the managed-care organization. As Duggan (2004) documents for U.S. Medicaid HMOs, and as Aizer et al. (2007) and Marton et al. (2014) discuss, this shift in residual cost-bearing creates an incentive to economize on input use, so the capitation margin reinforces the ownership-margin prediction in (5): $\Delta L^{\text{capitation}} \leq 0$.

Access component. The access margin combines two opposing forces. Extending formal coverage to the low-income population would, on its own, raise demand for insured care and predict an expansion of the local healthcare labor market ($\Delta L^{\text{access}} > 0$). Against it, the reform withdrew universal access at public facilities and replaced it with an income-based card that reached only part of the eligible (Finkelstein, Hendren, & Shepard, 2019; Finkelstein & Notowidigdo, 2019), confining enrollees to a single network with primary-care gatekeeping that rations utilization ($\Delta L^{\text{access}} \leq 0$). Because the pre-reform system already provided universal point-of-service access, the demand-expansion force is muted while the cost-containing features bind, so the access margin contracts the sector on net, $\Delta L^{\text{access}} \leq 0$.

Mobility margin. The reform’s labor-supply incidence is shaped by a feature distinctive to the Puerto Rican setting: the affected provider workforce holds U.S. citizenship and U.S.-credentialed professional licenses, so the legal cost of relocation to the mainland is zero. The

framework therefore admits an additional adjustment margin closed off in the mainland-U.S. place-based literature (Autor, Dorn, & Hanson, 2013; Yagan, 2019): out-migration of the displaced workforce. The empirical analysis traces this margin through the demographic-balancing identity (4), jointly with the level of county population. The two demographic measures are reported together because each is informative about a distinct aspect of the local labor supply, and disagreement between them carries identification content rather than being averaged away.

A.12 Empirical Strategy: Technical Detail

This subsection reports the dynamic-aggregation formula and the TWFE comparator specification referenced in Section 5.

Dynamic event-study aggregation. The event-study figures aggregate the group-time treatment effects $\widehat{\text{ATT}}(g, t)$ from (2) into event-time coefficients $\hat{\theta}_\ell$ by cohort-size weights:

$$\hat{\theta}_\ell = \sum_{g \in \mathcal{G}} \frac{N_g}{\sum_{g'} N_{g'}} \widehat{\text{ATT}}(g, g + \ell), \quad \ell \in \{-4, \dots, +4\}, \quad (6)$$

where N_g is the number of counties in cohort g and the reference period is $\ell = -1$. Confidence bands are constructed via the multiplier bootstrap of Callaway and Sant’Anna (2021) clustered at the county level with 1,000 replications, applied separately at each event time.

TWFE comparator specification. The canonical two-way fixed-effects (TWFE) event-study reported as estimator-class robustness in Appendix 10 regresses $Y_{i,t}$ on county fixed effects α_i , year fixed effects λ_t , and event-time indicators relative to treatment:

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{\ell \neq -1} \beta_\ell \mathbf{1}\{t - G_i = \ell\} + \varepsilon_{i,t}, \quad (7)$$

with standard errors clustered at the county level and the reference period $\ell = -1$ omitted.

Doubly robust three-step procedure. The $\widehat{\text{ATT}}(g, t)$ estimator in (2) is the doubly robust difference-in-differences estimator of Callaway and Sant’Anna (2021), which combines an inverse-probability-weighted estimator on the not-yet-treated comparison group with an outcome-regression-based correction term. The estimator is consistent if either the propensity-score model or the outcome-regression model is correctly specified, providing a robustness margin over either component alone. Implementation uses the `did` package in R with the default propensity-score specification (logistic regression on a constant) and the default outcome-regression specification (county and year fixed effects); the group-time-by-event-time estimates are then aggregated via (6) for the event-study figures and via the post-treatment-cell-count-weighted average for the regression tables. The not-yet-treated control group is implemented as `control_group = "notyettreated"` in `did::att_gt`.

A.13 BRFSS Coverage by Subgroup

This subsection expands on the BRFSS access-component result reported in Section 6.4. The BRFSS 1996–2000 panel restricts the analysis to Cohorts 3–5, for which a pre-treatment observation window exists within the panel (Appendix 10). The single-cohort exposure pattern within the BRFSS window means that Callaway and Sant’Anna (2021) is not implementable on this sample; the BRFSS results are reported under the Wooldridge (2025) ETWFE estimator with TWFE comparator estimates also reported. Six demographic subgroups are analyzed (Online Appendix Tables OA.14–OA.19): ages 18–29, 30–44, 45–64, and 65 and older; college graduates; and employed respondents. Coverage rises across all six subgroups, with the largest point estimate of +10.5 percentage points in the 45–64 age group ($p < 0.01$) and the smallest of +2.8 percentage points among respondents aged 65 and older ($p < 0.01$, smaller in magnitude because near-universal Medicare coverage at this age leaves limited room for the reform to expand effective insurance). The six subgroup estimates jointly rule out the no-coverage-change null even with the limited cohort exposure in the BRFSS sample, and the cross-subgroup pattern is consistent with the reform raising formal insurance coverage among the working-age low-income enrollees most directly targeted by its eligibility criteria.

A.14 CMS Hospital Cost Structure

As descriptive context for the hospital-capacity contraction in Table 5, Appendix Table A.8 reports characteristics of Puerto Rico’s Medicare-certified hospitals across the rollout, drawn from the CMS Inpatient Prospective Payment System (IPPS) Impact File merged into the Provider of Services panel (Centers for Medicare and Medicaid Services, 2023b). The surviving hospitals’ wage index held near 0.45 to 0.49 throughout, their case-mix index rose modestly from about 1.15 to 1.28, their Medicare share of inpatient days drifted up from about 0.43 toward 0.48, and their disproportionate-share measure changed little, so the contraction in the bed and service counts coincided with only a gradual shift in the cost-and-case-mix profile among the hospitals that remained certified. We read these series as suggestive evidence on hospital cost structure rather than as an identified effect of the reform; the file covers only Medicare-certified facilities and begins Puerto Rico coverage around 1992.

A.15 Cause-Specific Mortality

Appendix Table A.9 reports the cause-specific decomposition discussed in Section 6.4. We read it as a suggestive decomposition rather than a set of separately identified effects, by cause and at the horizon each permits. The margin our four-year window reads is case-fatality, the probability that a health event ends in death, which responds to local acute-care capacity within a few years; chronic-disease incidence responds over a far longer horizon. Cardiovascular and acute-trauma deaths are therefore informative on the case-fatality margin, where heart-disease and cerebrovascular mortality fall, while diabetes and pneumonia or influenza, which would move only through incidence, are unchanged and we do not read them. The homicide estimate we treat with particular caution, since its count blends vio-

Table A.8: Appendix — Suggestive Evidence from the CMS IPPS Impact File for PR Hospitals (1990–2002)

	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002
<i>IPPS Impact Fileⁱ</i>													
With Impact Data	0	0	0	55	55	54	53	53	53	53	52	50	51
Treated ^k	0	0	0	–	3	9	27	27	39	39	52	50	51
Control ^k	0	0	0	55	52	45	26	26	14	14	–	–	–
Wage Index	–	–	–	0.478	0.504	0.440	0.456	0.446	0.456	0.457	0.465	0.466	0.489
	(–)	(–)	(–)	(0.028)	(0.031)	(0.015)	(0.013)	(0.018)	(0.021)	(0.022)	(0.019)	(0.020)	(0.013)
Case Mix Index	–	–	–	1.149	1.171	1.206	1.231	1.243	1.249	1.256	1.262	1.280	1.258
	(–)	(–)	(–)	(0.172)	(0.188)	(0.314)	(0.270)	(0.257)	(0.236)	(0.276)	(0.282)	(0.293)	(0.269)
DSH Patient Share	–	–	–	–	0.126	0.142	0.141	0.133	0.137	0.117	0.123	0.127	0.124
	(–)	(–)	(–)	(–)	(0.209)	(0.222)	(0.216)	(0.211)	(0.210)	(0.176)	(0.178)	(0.187)	(0.165)
Medicare Utilization	–	–	–	–	0.432	0.419	0.458	0.462	0.481	0.492	0.486	0.484	0.471
	(–)	(–)	(–)	(–)	(0.157)	(0.126)	(0.136)	(0.119)	(0.152)	(0.132)	(0.106)	(0.100)	(0.115)
Avg Daily Census	–	–	–	–	92.0	107.2	107.3	108.4	96.7	108.2	98.2	101.4	99.9
	(–)	(–)	(–)	(–)	(70.2)	(75.8)	(77.3)	(75.8)	(75.8)	(77.4)	(77.3)	(80.3)	(79.7)
Beds (cost report)	–	–	–	–	172.3	159.6	153.2	153.9	161.2	152.7	157.5	160.7	156.3
	(–)	(–)	(–)	(–)	(114.8)	(96.2)	(95.7)	(93.2)	(94.1)	(87.7)	(91.7)	(92.9)	(93.7)

Notes: Means with standard deviations in parentheses. Hospital-year level, PR municipios only. Source: CMS IPPS Impact File, merged into the CMS Provider of Services provider panel by CMS_clean.R (Section 9 *IPPS Impact File merge*). The panel filters to PR-state providers (CMS provider numbers beginning with 40); each hospital-year is included if it has a non-NA wage index in the IPPS file for that year. This table presents suggestive evidence on hospital cost structure, case mix, and Medicare utilization; it is not a Main Figure outcome and is not interpreted causally.

ⁱ *Variable definitions:* Wage Index (IPPS `wageindex`) is the geographic adjustment factor applied to the labor-related share of IPPS base payment rates; values <1 indicate below-average local wages. Case Mix Index (IPPS `tacmiv`, total adjusted) is the average DRG weight across Medicare discharges; ≈ 1 is the national average. DSH Patient Share (IPPS `dshpct`) is the Disproportionate Share Hospital adjustment ratio (low-income Medicare + Medicaid days / total days), reported as a fraction. Medicare Utilization (IPPS `mcr_pct`) is the share of inpatient days that are Medicare, reported as a fraction. Avg Daily Census (`adc`) and Beds (cost report; `beds`) come from the Medicare cost report.

^k *Treated/Control:* A hospital-year is Treated if its municipio's `reforma_cohort` year is at or before the column year (matching the cumulative-treatment definition used in Table 4). Hospitals in untreated municipios, or treated municipios with `reforma_cohort` year strictly after the column year, are Control. Cells are – when no PR hospital appears in the IPPS Impact File for that year (the file begins PR coverage in ≈ 1992). Three additional IPPS variables from initial drafts (Cost-to-Charge Ratio, Resident/ADC Ratio, Medicare Discharges) are not in CMS_clean.R's current IPPS extractor and are omitted here.

lence incidence, which rose across Puerto Rico through the 1990s for reasons unrelated to the health system, with the case-fatality margin emergency care governs. Maternal mortality is the least readable of the movements: it rests on a marginally significant estimate off a very small pre-reform base and on care managed over a longer horizon than our window, so we read its increase as suggestive only. This is why we take the infant mortality rate as the primary patient-health outcome, since newborn survival turns on immediate, high-technology intervention and is sensitive to short-term, localized care quality (Cutler & Meara, 2000; Cutler, Rosen, & Vijan, 2006).

Table A.9: Cause-Specific Mortality Effects of The Reform

	(1) Heart	(2) Cerebro.	(3) Diabetes	(4) Pneum./Flu	(5) Perinatal	(6) Maternal	(7) Accidents	(8) Homicide
ATT	-0.096** (0.046)	-0.041** (0.020)	-0.010 (0.034)	0.009 (0.021)	–	0.002* (0.001)	0.138 (0.101)	0.107*** (0.015)
Pre-treatment mean of dep. var.	1.572	0.370	0.524	0.298	–	0.003	0.344	0.143
ATT as % of pre-treatment mean	-6.1%	-11.0%	-2.0%	3.1%	–	73.1%	40.1%	75.0%
<i>N</i> counties	78	78	78	78	–	78	78	78

Notes: Estimates use the Callaway and Sant’Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. **Outcomes.** Cause-specific death rates per 1,000 residents. **Caveat.**

Online Appendix

The Online Appendix collects the additional sample-restriction, estimator-comparison, and subgroup specifications referenced in the main text. The exhibits are organized by topic, in the order in which they are referenced in the body.

OA.1 Macro Labor-Market Outcomes

Reports the BLS-standard unemployment rate, the labor-force-participation rate, and the employment-to-population rate side by side. The unemployment rate is statistically indistinguishable from zero against a pre-reform mean of 17.770 percent; the LFP and employment-to-population rates both fall by 3.5 percent of pre-mean. The unemployment-rate null together with the LFP decline is the labor-supply-margin signature of Section 6.3.

OA.2 Log-Specification on QCEW Main

Parallel specification on $\log(1+y)$ per-1,000 normalizations and $\log(1+y)$ raw-level outcomes for the QCEW healthcare aggregate. The per-capita and raw-level estimates point in opposite directions because the local population grew faster than the local healthcare workforce during the analysis window; the main finding rests on the per-capita normalization.

OA.3 Excluding San Juan: First-Stage and Patient Health

Reports the side-by-side balanced and excluded-San-Juan estimates for the government-hospital birth share, the infant mortality rate, and the low-birth-weight share. The first-stage

Table OA.1: Macro Labor-Market Effects of the reform (Online Appendix)

	(1) Unemployment Rate (%)	(2) Employment Rate (%)	(3) Labor Force Rate (%)
ATT	-0.245 (0.285)	-0.926*** (0.219)	-1.110*** (0.241)
Pre-treatment mean of dep. var.	17.770	26.331	31.895
ATT as % of pre-treatment mean	-1.4%	-3.5%	-3.5%
counties	78	78	78

Notes: Estimates use the Callaway and Sant'Anna (2021) ATT. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors.

Table OA.2: Log(1+x) Specification: QCEW Healthcare Headline Outcomes (Online Appendix)

	Per-1k Population (log1p)			Levels (log1p)		
	(1) Empl./1k	(2) Wages/Empl./1k	(3) Estabs./1k	(4) Empl.	(5) Wages	(6) Estabs.
ATT	0.0816*** (0.0274)	-0.0180 (0.1137)	0.0176* (0.0102)	0.1902** (0.0743)	0.2469 (0.2635)	0.1438*** (0.0482)
Pre-treatment mean of dep. var.	1.2007	3.2910	0.5134	4.2928	11.0005	3.1613
ATT as % of pre-treatment mean	6.8%	-0.5%	3.4%	4.4%	2.2%	4.5%
counties	59	59	59	59	59	59

Notes: Estimates use the Callaway and Sant'Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors.

ownership-share estimate moves from -14.79 to -8.22 percentage points (both $p < 0.01$); the patient-health outcomes are robust to the exclusion.

Table OA.3: Excluding San Juan: Patient-Health and Birth-Share Robustness (Online Appendix)

	Gov. Hosp. Birth Share		IMR		Low BW Rate	
	(1) Bal.	(2) Excl. SJ	(3) Bal.	(4) Excl. SJ	(5) Bal.	(6) Excl. SJ
ATT	-14.79*** (1.94)	-8.22*** (2.85)	-1.205 (0.928)	0.781 (1.237)	0.060 (0.230)	-0.307 (0.368)
Pre-treatment mean of dep. var.	62.55	62.61	12.620	12.624	9.717	9.712
ATT as % of pre-treatment mean	-23.6%	-13.1%	-9.5%	6.2%	0.6%	-3.2%
counties	78	77	78	77	78	77

Notes: Estimates use the Callaway and Sant'Anna (2021) doubly-robust ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors.

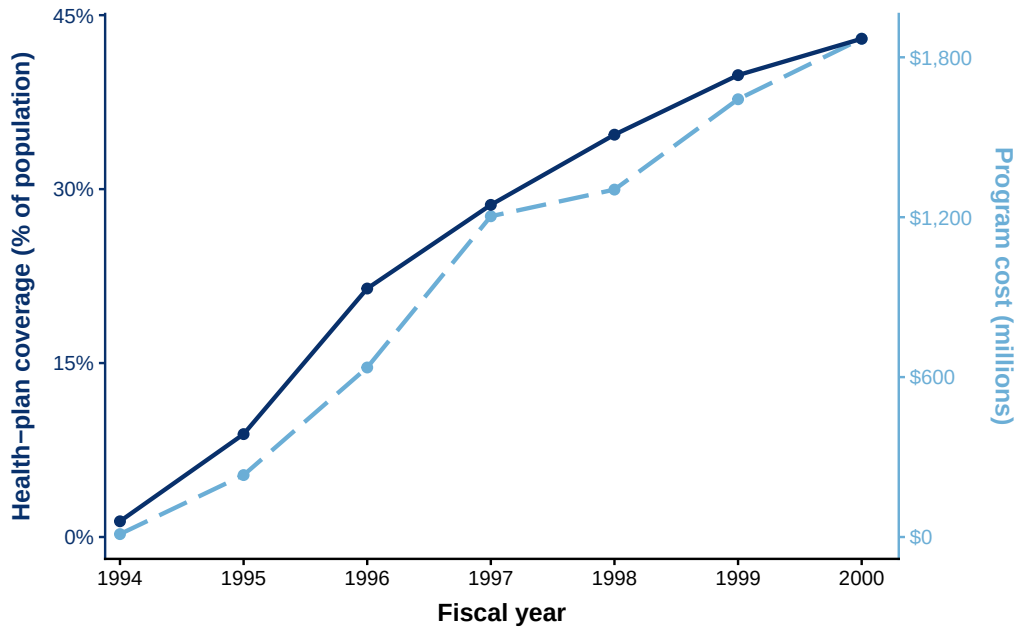


Figure A.3: **ASES coverage and program cost, 1994–2000.** The share of Puerto Rico residents enrolled in the ASES health card (left axis; enrollees divided by the VE time-varying island population, the paper’s denominator) and the program’s cost in constant 2025 US dollars (right axis; deflated with the U.S. CPI-U). Enrollment ramped from a pilot to a large share of the population as program cost scaled to about \$1.0 billion in nominal terms, near \$1.9 billion in the constant 2025 dollars of the right axis. The plotted enrollment line is read from the published budget figure and is approximate; the enrollment counts of record are the ASES administrative tabulation (Administración de Seguros de Salud de Puerto Rico, 2026), which reports 1,795,067 insured residents by 2000 (about 47 percent of the population and 96 percent of the eligible) and supersedes the chart-read. The cost series cross-checks the independently compiled by-agency ASES budget to within rounding at every overlapping year. The figure is descriptive and does not identify an effect of the reform. Sources: Administración de Seguros de Salud de Puerto Rico (2026) for enrollment and Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto (1999) for cost, with the VE population denominator.

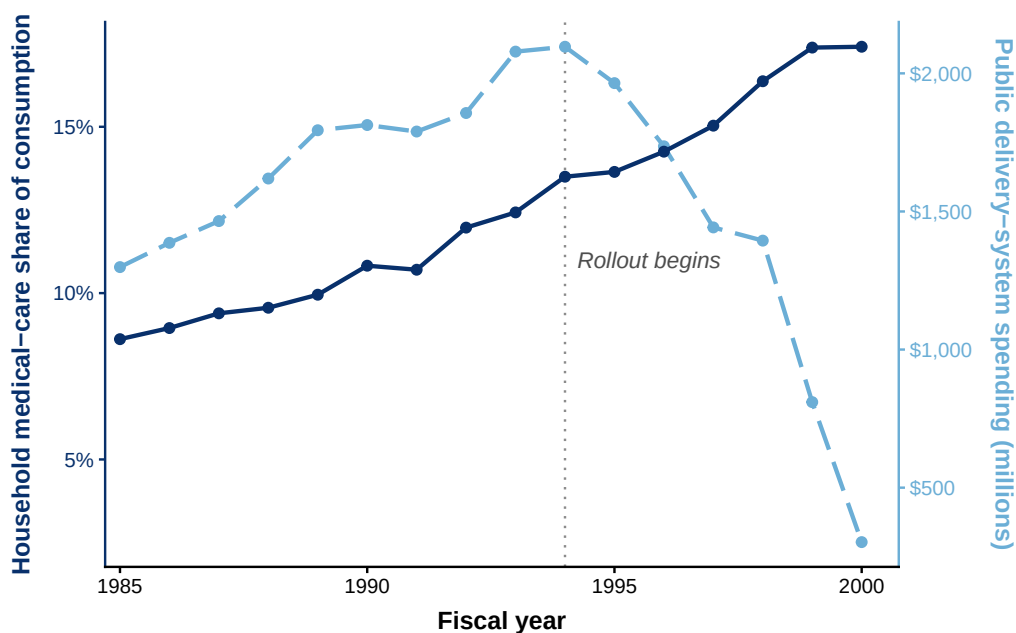


Figure A.4: **Consumer medical spending and public delivery spending, 1985–2000.** The household medical-care expenditure share of total personal consumption (left axis, solid navy line) against spending on the public delivery agencies (the Assistance Fund for Health Services, the Medical Center, and the Cardiovascular Center; right axis, dashed orange line). The two rise together while the public sector still operated delivery and diverge over the rollout, as public delivery spending fell and the household share continued upward. The household series is a national-accounts personal-consumption share rather than out-of-pocket spending, and it had been rising for two decades before the reform, from 5.6 percent in 1971 to 12.4 percent in 1993, so the figure is descriptive and does not attribute the household trend to the reform. The public delivery series is reported on a financial-statements basis below the operating-budget totals used elsewhere. Public delivery-system spending is expressed in constant 2025 US dollars, deflated with the U.S. CPI-U; the household series is a consumption share (a ratio) and is not deflated. Sources: Junta de Planificación de Puerto Rico (2024) (Table 5, Personal Consumption Expenditures) and Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto (1999).

OA.4 Excluding San Juan: CMS Provider-Stratified

Reports excluded-San-Juan robustness for CMS hospital-bed and services-per-thousand outcomes, stratified by ownership class.

Table OA.4: Excluding San Juan: CMS Provider-Structure Robustness (Online Appendix)

	Beds per 1k pop.		Services per 1k pop.		FTE per 1k pop.	
	(1)	(2)	(3)	(4)	(5)	(6)
	Bal.	Excl. SJ	Bal.	Excl. SJ	Bal.	Excl. SJ
ATT	-0.265*** (0.065)	0.090 (0.140)	-0.024** (0.010)	-0.014 (0.012)	-2.251** (0.971)	-1.815 (1.380)
Pre-treatment mean of dep. var.	3.389	3.033	0.487	0.463	10.828	10.288
ATT as % of pre-treatment mean	-7.8%	3.0%	-4.9%	-3.0%	-20.8%	-17.6%
counties	40	39	40	39	33	32

Notes: Estimates use the Callaway and Sant’Anna (2021) doubly-robust ATT estimator with not-yet-treated comparison units. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors.

OA.5 Spillover-Excluded Sample: First Stage and Patient Health

The first-stage ownership-share estimate on the non-border subsample is -12.14 percentage points ($p < 0.01$, -19.4 percent of pre-mean), economically indistinguishable from the main result; the corresponding patient-health estimates are statistically indistinguishable from zero on this restricted panel.

Table OA.5: Spillover-Excluded Sample: Hospital Birth Share and Patient-Health Outcomes

	Hospital Birth Share (VS)		Patient Health (VS)	
	(1)	(2)	(3)	(4)
	Gov. Hosp. (%)	Priv. Hosp. (%)	IMR (per 1k)	Low BW (%)
ATT	-12.14*** (2.68)	12.29*** (2.44)	-0.072 (1.508)	0.087 (0.578)
Pre-treatment mean of dep. var.	62.48	37.00	12.534	9.524
ATT as % of pre-treatment mean	-19.4%	33.2%	-0.6%	0.9%
counties	39	39	39	39

Notes: Estimates use the Callaway and Sant’Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors.

OA.6 Spillover-Excluded Sample: CMS Provider Capacity

CMS hospital-bed and services estimates on the spatially-restricted, SUTVA-tightened sample.

Table OA.6: Spillover-Excluded Sample: CMS Provider-Structure Outcomes

	(1) Beds per 1k pop.	(2) Services per 1k pop.	(3) FTE per 1k pop.
ATT	-0.003 (0.061)	-0.020 (0.023)	-3.278 (2.282)
Pre-treatment mean of dep. var.	3.908	0.657	14.272
ATT as % of pre-treatment mean	-0.1%	-3.1%	-23.0%
<i>N</i> counties	19	19	16

Notes: Estimates use the Callaway and Sant’Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors.

OA.7 Low-Commuting-Inflow Sample: First Stage and Patient Health

Sample restricted to counties with workplace in-commuting share at or below 0.50 ($n = 64$). The first-stage ownership-share estimate is -7.18 percentage points ($p < 0.05$); the patient-health estimates are robust. Therefore, if we assume that in low-commute counties treatment-effect contamination through cross-county worker flows is minimal, then may assume that our estimates are unbiased, since the effects on the patient-health margin remain consistent with the main estimates.

Table OA.7: Low-Commuting-Inflow Sample: Hospital Birth Share and Patient Health

	(1) Gov. Hosp. Birth Share (%)	(2) IMR (per 1k live births)
ATT	-7.18** (3.29)	1.327 (1.392)
Pre-treatment mean of dep. var.	64.11	12.846
ATT as % of pre-treatment mean	-11.2%	10.3%
<i>N</i> counties	64	64

Notes: Estimates use the Callaway and Sant’Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. Sample restricted to counties with *low* workplace in-commuting share ($\text{in_share} \leq 0.50$, where in_share = workers commuting in from other counties divided by total workplace count). The classifier is built from Census 2000 Journey-to-Work tabulations and read from the diagnostic CSV that `all_event_studies.R` writes at startup.

OA.8 Low-Commuting-Inflow Sample: CMS Provider Capacity

CMS hospital-bed and services estimates on the low-commuting-inflow subsample. CMS supply-side responses in low-commute counties are consistent with main estimates.

Table OA.8: Low-Commuting-Inflow Sample: CMS Provider-Structure Outcomes

	(1) Beds per 1k pop.	(2) Services per 1k pop.	(3) FTE per 1k pop.
ATT	0.087 (0.174)	-0.007 (0.011)	-2.113 (1.567)
Pre-treatment mean of dep. var.	2.746	0.415	9.730
ATT as % of pre-treatment mean	3.2%	-1.7%	-21.7%
<i>N</i> counties	34	34	27

Notes: Estimates use the Callaway and Sant’Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors.

OA.9 Heterogeneity by Pre-Reform Population

The infant-mortality estimate in the above-median (larger) cell is -2.564 per thousand live births ($p < 0.01$, -20.8 percent of pre-mean). Second of the four heterogeneity dimensions on which the IMR signal is stronger than the baseline-exposure cut.

Table OA.9: Heterogeneity by Pre-Reform Population (1990–1993 mean) (Online Appendix)

	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
<i>Below-median pre-reform population</i>				
ATT	-0.37 (5.78)	-0.154 (0.221)	-1.232 (11.833)	3.657* (2.081)
Pre-treatment mean	66.04	1.496	0.050	12.923
ATT as % of pre-mean	-0.6%	-10.3%	-2464.3%	28.3%
<i>N</i> counties	39	20	39	39
<i>Above-median pre-reform population</i>				
ATT	-17.38*** (2.27)	-0.300 (0.228)	-14.651 (11.447)	-2.564*** (0.921)
Pre-treatment mean	59.33	5.085	1.975	12.347
ATT as % of pre-mean	-29.3%	-5.9%	-741.7%	-20.8%
<i>N</i> counties	39	39	39	39

Notes: Estimates use the Callaway and Sant’Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. The classifier is computed once over the strict pre-rollout window (BASELINE_YEARS = 1990–1993) and is therefore time-invariant within county. Counties without a valid baseline value are routed to the below-median bin per the standard censoring rule. *Below-median* = 39 counties; *Above-median* = 39 counties.

OA.10 Heterogeneity by Pre-Reform Healthcare Employment

The infant-mortality estimate in the above-median (higher-pre-reform-healthcare-intensity) cell is -2.633 per thousand live births ($p < 0.01$, -20.7 percent of pre-mean). Third of the four heterogeneity dimensions on which the IMR signal is stronger than the baseline-exposure cut.

Table OA.10: Heterogeneity by Pre-Reform HC Employment (1990–1993 mean)

	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
<i>Below-median pre-reform HC employment</i>				
ATT	-4.69 (3.68)	-0.084 (0.196)	-8.171 (11.155)	2.503 (1.634)
Pre-treatment mean	64.79	1.136	1.440	12.572
ATT as % of pre-mean	-7.2%	-7.4%	-567.5%	19.9%
<i>N</i> counties	48	29	48	48
<i>Above-median pre-reform HC employment</i>				
ATT	-17.87*** (2.54)	-0.433 (0.268)	-7.758 (13.732)	-2.633*** (0.830)
Pre-treatment mean	58.85	6.816	0.428	12.698
ATT as % of pre-mean	-30.4%	-6.4%	-1810.6%	-20.7%
<i>N</i> counties	30	30	30	30

Notes: Estimates use the Callaway and Sant’Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. Counties are partitioned at the cross-county median of the 1990–1993 mean QCEW Healthcare (NAICS 621+622+623) employment level (annualized from quarterly cells with ≥ 3 valid quarters) (median = 58). The classifier is computed once over the strict pre-rollout window (BASELINE.YEARS = 1990–1993) and is therefore time-invariant within county. Counties without a valid baseline value are routed to the below-median bin. *Below-median* = 48 counties; *Above-median* = 30 counties.

OA.11 Heterogeneity by Workplace Commuting Inflow

The infant-mortality estimate in the above-median (workplace-center) cell is -3.124 per thousand live births ($p < 0.01$, -26.8 percent of pre-mean). Fourth of the four heterogeneity dimensions on which the IMR signal is stronger than the baseline-exposure cut.

Table OA.11: Heterogeneity by Workplace Commuting Inflow (Low vs. High Inflow)

	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
<i>Low-inflow workplace counties</i>				
ATT	-7.18** (3.29)	-0.356 (0.218)	-4.777 (9.854)	1.327 (1.392)
Pre-treatment mean	64.11	2.910	1.916	12.846
ATT as % of pre-mean	-11.2%	-12.3%	-249.4%	10.3%
<i>N</i> counties	64	45	64	64
<i>High-inflow workplace counties</i>				
ATT	-16.55*** (3.02)	-0.021 (0.357)	7.774 (11.287)	-3.124*** (1.114)
Pre-treatment mean	55.99	6.974	-2.495	11.671
ATT as % of pre-mean	-29.6%	-0.3%	-311.6%	-26.8%
<i>N</i> counties	14	14	14	14

Notes: Estimates use the Callaway and Sant’Anna (2021) doubly-robust ATT estimator. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. Workplace in-commuting share from Census 2000 Journey-to-Work tabulations: $\text{in_share} = (\text{in-commuters from other counties}) / (\text{total workplace count})$. Threshold cutoff at 0.50 (NOT median). *Low-inflow* = 45 counties ($\text{in_share} \leq 0.50$); *High-inflow* = 14 counties ($\text{in_share} > 0.50$). **Note.** The ‘Below-median’ and ‘Above-median’ panels here label the low-inflow and high-inflow groups respectively. **Reading.** High-inflow workplace counties import labor-force flows from neighbors; if treatment effects differ across the inflow margin, that’s evidence of cross-county worker spillovers acting on the labor-market response.

OA.12 Per-Thousand-Resident Placebo Industries

C&S estimates on three non-healthcare placebo industries: NAICS 11 (Agriculture, Forestry, and Fishing), NAICS 48–49 (Transportation and Warehousing), and NAICS 71 (Arts, Entertainment, and Recreation). NAICS 31–33 manufacturing is excluded *a priori* as endogenous to the contemporaneous §936 corporate-tax phase-out.

Table OA.12: QCEW Placebo Industries: Effects of the reform on Industries Outside the Healthcare Sector

Industry	(1) Empl. per 1k	(2) Wages/Empl per 1k	(3) Wages per 1k	(4) Estabs per 1k
<i>Agriculture, Forestry & Fishing (NAICS 11)</i>				
ATT	0.108 (0.369)	-6.126*** (1.591)	-1,476.474** (741.852)	0.106 (0.075)
Pre-treatment mean	12.463	30.797	—	3.463
ATT as % of pre-mean	0.9%	-19.9%	—	3.1%
counties	25	25		25
<i>Transportation & Warehousing (NAICS 48-49)</i>				
ATT	-0.709*** (0.046)	7.571 (9.813)	-688.266*** (243.595)	0.001 (0.003)
Pre-treatment mean	1.475	240.942	—	0.105
ATT as % of pre-mean	-48.1%	3.1%	—	1.2%
counties	77	77		77
<i>Arts, Entertainment & Recreation (NAICS 71)</i>				
ATT	-0.498 (0.350)	1.994 (1.329)	-553.871 (558.662)	-0.032** (0.015)
Pre-treatment mean	2.107	15.207	—	0.155
ATT as % of pre-mean	-23.6%	13.1%	—	-20.6%
counties	9	9		9

Notes: Estimates use the Callaway and Sant'Anna (2021). Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. **Sample.** QCEW (BLS) county-quarter panel restricted to one of three placebo industries: NAICS 11 (Agriculture, Forestry, Fishing & Hunting); NAICS 48-49 (Transportation & Warehousing); NAICS 71 (Arts, Entertainment & Recreation).

OA.13 Raw-Level Placebo Industries

Raw-level (rather than per-thousand-resident) specifications on the same three placebo industries. Agriculture is null in raw levels, supporting the per-capita-redistribution interpretation of the per-capita Agriculture-wage-bill decline reported in Section 7. Transportation contracts in raw levels by 94.0 percent, consistent with input-output linkage to the phasing-out manufacturing logistics chain. Arts is statistically underpowered. For context, a per-1k placebo null could in principle hide an offsetting population-denominator shock that the per-1k normalization absorbs. Raw-level placebos provide a cleaner test for absolute employment-flow effects.

Table OA.13: QCEW Placebo Industries (Raw Levels): Employment, Wage Bill, and Establishment Counts

Industry	(1) Employment	(2) Wage Bill	(3) Establishments
<i>Agriculture, Forestry & Fishing (NAICS 11)</i>			
ATT	14.5 (13.8)	12,717 (20,913)	0.3 (2.0)
Pre-treatment mean	293.0	193,903	85.5
ATT as % of pre-mean	4.9%	6.6%	0.3%
<i>N</i> counties	25	25	25
<i>Transportation & Warehousing (NAICS 48-49)</i>			
ATT	-191.9*** (11.8)	-164,320** (83,737)	1.2*** (0.3)
Pre-treatment mean	204.1	791,798	7.4
ATT as % of pre-mean	-94.0%	-20.8%	15.9%
<i>N</i> counties	77	77	77
<i>Arts, Entertainment & Recreation (NAICS 71)</i>			
ATT	-63.7 (51.5)	-192,039* (106,043)	-14.4*** (3.8)
Pre-treatment mean	309.2	568,279	30.1
ATT as % of pre-mean	-20.6%	-33.8%	-47.7%
<i>N</i> counties	9	9	9

Notes: Three estimators are reported. TWFE: two-way fixed-effects regression with county and year FEs and event-time dummies; the overall ATT is the simple mean of the post-treatment dummy coefficients with delta-method standard error on the post-period. ETWFE: Wooldridge (2021) Extended Two-Way Fixed Effects, with overall ATT as a cohort-size-weighted average. C&S: Callaway and Sant'Anna (2021) doubly-robust ATT, aggregated to a single overall ATT. All three estimators use cluster-robust standard errors at the county level; C&S uses the multiplier bootstrap (1000 replications). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. Raw-level outcomes: Employment is the average quarterly headcount; Wage Bill is the average quarterly real wage bill (deflated; see Section 12b for the deflator); Establishments is the average quarterly establishment count.

OA.14 BRFSS Insurance Coverage by Subgroup

BRFSS subgroup estimates of the ETWFE point estimate on the “has health plan” indicator. The formal-insurance response is positive and significant in five of the six reported subgroups; the 30–44 age group is the one in which the estimate is positive but does not reach conventional significance. The pattern is consistent with the BRFSS analysis covering Cohorts 3–5 (the cohorts for which a pre-treatment BRFSS observation window is available), documented in Appendix 10.

Table OA.14: BRFSS Healthcare Access — Age 18–29

	(1) Has Health Plan	(2) Preventive Services	(3) Could Not Afford Doctor	(4) General Health (1-5)
TWFE (weighted)	-0.0339 (0.0253)	-0.215*** (0.033)	-0.0109 (0.0152)	0.256*** (0.049)
ETWFE	0.0783*** (0.0232)	-0.109* (0.056)	-0.0118 (0.0245)	-0.001 (0.052)

Notes: TWFE estimated with BRFSS survey weights (`weights_var = 'survey_weight'`); ETWFE is unweighted. Cells report ATT (SE in parentheses). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Callaway-Sant’Anna is not run for BRFSS because the dataset’s 1996–2000 window contains only one fully-treated reform cohort (1998); the 2000 cohort has no valid not-yet-treated controls in the post-period. Sample: BRFSS respondents aged 18–29. Aggregated from `all_brfss_coefs.csv` produced by `all_event_studies.R`; same numerical conventions as Online Appendix Figure OA.15.

Table OA.15: BRFSS Healthcare Access — Age 30–44

	(1) Has Health Plan	(2) Preventive Services	(3) Could Not Afford Doctor	(4) General Health (1-5)
TWFE (weighted)	-0.0053 (0.0145)	0.094*** (0.026)	0.0190** (0.0082)	-0.040 (0.035)
ETWFE	0.0301 (0.0186)	0.025 (0.029)	0.0147 (0.0174)	-0.107** (0.044)

Notes: TWFE estimated with BRFSS survey weights (`weights_var = 'survey_weight'`); ETWFE is unweighted. Cells report ATT (SE in parentheses). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Callaway-Sant’Anna is not run for BRFSS because the dataset’s 1996–2000 window contains only one fully-treated reform cohort (1998); the 2000 cohort has no valid not-yet-treated controls in the post-period. Sample: BRFSS respondents aged 30–44. Aggregated from `all_brfss_coefs.csv` produced by `all_event_studies.R`; same numerical conventions as Online Appendix Figure OA.16.

Table OA.16: BRFSS Healthcare Access — Age 45–64

	(1) Has Health Plan	(2) Preventive Services	(3) Could Not Afford Doctor	(4) General Health (1-5)
TWFE (weighted)	0.0891*** (0.0084)	0.133*** (0.033)	0.0117 (0.0101)	0.017 (0.040)
ETWFE	0.1052*** (0.0138)	0.100*** (0.032)	-0.0102 (0.0120)	0.102 (0.073)

Notes: TWFE estimated with BRFSS survey weights (`weights_var = 'survey_weight'`); ETWFE is unweighted. Cells report ATT (SE in parentheses). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Callaway-Sant’Anna is not run for BRFSS because the dataset’s 1996–2000 window contains only one fully-treated reform cohort (1998); the 2000 cohort has no valid not-yet-treated controls in the post-period. Sample: BRFSS respondents aged 45–64. Aggregated from `all_brfss_coefs.csv` produced by `all_event_studies.R`; same numerical conventions as Online Appendix Figure OA.17.

Table OA.17: BRFSS Healthcare Access — Age 65 and Older

	(1) Has Health Plan	(2) Preventive Services	(3) Could Not Afford Doctor	(4) General Health (1-5)
TWFE (weighted)	0.0393*** (0.0049)	0.037*** (0.012)	0.0001 (0.0127)	0.138*** (0.033)
ETWFE	0.0280*** (0.0087)	-0.023 (0.014)	-0.0101 (0.0144)	0.211*** (0.052)

Notes: TWFE estimated with BRFSS survey weights (`weights_var = 'survey_weight'`); ETWFE is unweighted. Cells report ATT (SE in parentheses). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Callaway-Sant'Anna is not run for BRFSS because the dataset's 1996–2000 window contains only one fully-treated reform cohort (1998); the 2000 cohort has no valid not-yet-treated controls in the post-period. Sample: BRFSS respondents aged 65+ (Medicare-eligible). La Reforma did not directly target this group; near-null on health-plan coverage is the expected no-effect benchmark. Aggregated from `all_brfss_coefs.csv` produced by `all_event_studies.R`; same numerical conventions as Online Appendix Figure OA.18.

Table OA.18: BRFSS Healthcare Access — College Graduates

	(1) Has Health Plan	(2) Preventive Services	(3) Could Not Afford Doctor	(4) General Health (1-5)
TWFE (weighted)	-0.0375*** (0.0104)	0.085*** (0.026)	0.0524*** (0.0103)	0.111*** (0.038)
ETWFE	0.0810*** (0.0212)	0.059* (0.032)	0.0305 (0.0209)	0.201*** (0.059)

Notes: TWFE estimated with BRFSS survey weights (`weights_var = 'survey_weight'`); ETWFE is unweighted. Cells report ATT (SE in parentheses). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Callaway-Sant'Anna is not run for BRFSS because the dataset's 1996–2000 window contains only one fully-treated reform cohort (1998); the 2000 cohort has no valid not-yet-treated controls in the post-period. Sample: BRFSS respondents with college degree. Aggregated from `all_brfss_coefs.csv` produced by `all_event_studies.R`; same numerical conventions as Online Appendix Figure OA.20.

Table OA.19: BRFSS Healthcare Access — Employed Respondents

	(1) Has Health Plan	(2) Preventive Services	(3) Could Not Afford Doctor	(4) General Health (1-5)
TWFE (weighted)	-0.0044 (0.0111)	0.029 (15,216.475)	0.0120 (0.0077)	-0.019 (0.034)
ETWFE	0.0460*** (0.0102)	0.016 (0.029)	-0.0233** (0.0104)	-0.144*** (0.037)

Notes: TWFE estimated with BRFSS survey weights (`weights_var = 'survey_weight'`); ETWFE is unweighted. Cells report ATT (SE in parentheses). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Callaway-Sant'Anna is not run for BRFSS because the dataset's 1996–2000 window contains only one fully-treated reform cohort (1998); the 2000 cohort has no valid not-yet-treated controls in the post-period. Sample: BRFSS respondents reporting employment. Aggregated from `all_brfss_coefs.csv` produced by `all_event_studies.R`; same numerical conventions as Online Appendix Figure OA.19.

OA.15 Welfare Bounds: Refinements

Assesses two refinements to the welfare bounds, applied jointly: (R1) the 0.5 high-substitution-half multiplier is replaced with the empirical 1990–1993 share of live births in the high-substitution subset, averaged across the four heterogeneity dimensions (51.9 percent); and (R2) the wage saving is scaled by a reabsorption factor $1 - f$ where f is the share of displaced healthcare workers who fail to reabsorb into productive employment. Under current data the unemployment-rate ATT (Table OA.1) is statistically indistinguishable from zero, so the reabsorption factor equals one and refinement (R2) does not move the cells. The empirical-births-share refinement (R1) approximately coincides with the 0.5 multiplier used in Table A.4, so the refined cells deviate from the original by less than 3 percent. We therefore treat the refinement as a robustness exercise rather than an alternative main result and omit a separate table.

OA.16 Population in Levels and Logarithms

Reports the C&S point estimate on total county population in level and log specifications. The level estimate is +5.1 percent of pre-mean ($p < 0.01$); the log specification is consistent in sign and magnitude. The population growth estimate is the level outcome that complements the demographic-balancing residual (CNMR) reported in Table 6.

Table OA.20: Effects of The Reform on Log County Population

	(1) Population (levels)	(2) Log Population
ATT	2,528*** (690)	0.0357*** (0.0086)
Pre-treatment mean of dep. var.	49,898	10.4172
ATT as % of pre-treatment mean	5.1%	0.3%
N counties	78	78

Notes: Estimates use the Callaway and Sant’Anna (2021) doubly-robust ATT estimator. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. Total county population (Col. 1) and its natural logarithm (Col. 2).

OA.17 Demographic Reconciliation

Reports the explicit demographic-balancing identity check between the population-level C&S point estimate and the sum of the implied contributions from natural increase (births minus deaths) and net migration. The residual of approximately 66 per thousand reflects (i) measurement error in the CNMR, which is constructed as a residual of three measured flows, and (ii) C&S aggregation weights that differ across panels with different observation counts (the CNMR loses each county’s first observation because Pop_{t-1} is unavailable). The two demographic margins are reported jointly because the disagreement carries identification content.

Table OA.21: Demographic Margins of Adjustment: Reconciling Levels and Flows

	Population (Levels)		Net Migration	Vital Rates (Flows)	
	(1) Population	(2) Log Population	(3) CNMR (per 1k)	(4) Crude Birth Rate (per 1k)	(5) Crude Death Rate (per 1k)
ATT	2,528*** (690)	0.0357*** (0.0086)	-14.651** (7.389)	-0.564** (0.225)	0.047 (0.114)
Pre-treatment mean of dep. var.	49,898	10.4172	1.059	18.267	7.293
ATT as % of pre-treatment mean	5.1%	0.3%	-1382.9%	-3.1%	0.6%
<i>N</i> counties	78	78	78	78	78

Notes: Estimates use the Callaway and Sant’Anna (2021) doubly-robust ATT estimator. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. The demographic identity for any closed area between two periods is $\Delta \text{Population}_{t,t-1} = (\text{Births}_t - \text{Deaths}_t) + \text{Net Migration}_t$, and dividing by average population yields the per-1k version: $\Delta \log P \approx \text{CBR} - \text{CDR} + \text{CNMR}$. Columns (3)–(5) report the C&S ATT for each of the three flow components on the right-hand side; columns (1)–(2) report the left-hand side in absolute and proportional units.

OA.18 NVSS Individual-Level Birth-Outcome Robustness

Reports C&S estimates on individual-level NVSS natality records using the conception-year-treatment indicator. Outcomes: birth weight, the count of prenatal visits, the trimester of prenatal-care initiation, and the five-minute Apgar score. The point estimates are statistically indistinguishable from zero, consistent with the county-aggregate Vital Statistics null reported in Table 5.

Table OA.22: NVSS Individual-Level Birth Outcomes — Effects of The Reform

	(1) Birth Weight (g)	(2) Prenatal Visits	(3) Apgar (5-min)	(4) Pr(LBW)	(5) Pr(VLBW)
ATT (C&S)	-0.75 (9.99)	-0.18 (0.17)	0.002 (0.020)	0.0015 (0.0036)	-0.0001 (0.0013)

Notes: Estimates use the Callaway and Sant’Anna (2021) doubly-robust ATT estimator. Cluster-robust standard errors at the county (municipio) level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. The variables are continuous birth-weight, prenatal-visits, and Apgar measures plus binary low/very-low-birth-weight indicators.

OA.19 Heterogeneity by Pre-Reform Poverty Rate

Partitions the seventy-eight counties at the median Census 2000 poverty rate (U.S. Census Bureau, 2000b). The healthcare-employment estimate in the above-median-poverty cell is -31.2 percent of pre-mean ($p < 0.10$); the result is directionally consistent with concentration of the contraction in slack-labor-market cells. Sample-split power limits formal inference on the access-component-versus-ownership-component decomposition contemplated by the framework of Section 3.

Table OA.23: Heterogeneity by Census 2000 Poverty Rate

	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
<i>Below-median poverty rate</i>				
ATT	-17.81*** (1.93)	-0.295 (0.214)	-15.773 (11.426)	-1.622* (0.984)
Pre-treatment mean	57.92	5.481	2.166	11.970
ATT as % of pre-mean	-30.7%	-5.4%	-728.1%	-13.5%
<i>N</i> counties	39	34	39	39
<i>Above-median poverty rate</i>				
ATT	-2.49 (4.76)	-0.557* (0.290)	1.249 (11.907)	2.066 (1.972)
Pre-treatment mean	67.14	1.787	-0.037	13.265
ATT as % of pre-mean	-3.7%	-31.2%	-3378.7%	15.6%
<i>N</i> counties	39	25	39	39

Notes: Estimates use the Callaway and Sant'Anna (2021) doubly-robust ATT estimator. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. Counties are partitioned at the cross-county median of the Census 2000 % of residents below the federal poverty line (median = 54.55%). The classifier is computed once over the strict pre-rollout window (baseline years = 1990–1993) and is therefore time-invariant within county. Counties without a valid baseline value are routed to the below-median bin per the standard censoring rule. *Below-median* = 39 counties; *Above-median* = 39 counties. Census 2000 is partially post-rollout for cohorts 1–4 (1994–1998) and concurrent with cohort 5 (2000), so the classifier is partially endogenous to the reform itself; yet we retain it because no comparable pre-reform county-level poverty series exists.

OA.20 Heterogeneity by Pre-Reform Unemployment Rate

Partitions the seventy-eight counties at the median pre-reform unemployment rate. The healthcare-employment estimate in the above-median-unemployment cell is -17.8 percent of pre-mean (statistically insignificant); the directional pattern is consistent with the poverty-rate cut.

Table OA.24: Heterogeneity by Pre-Reform Unemployment Rate (BLS, 1990–1993 mean)

	(1) Gov. Hosp. Birth Share (%)	(2) HC Empl. per 1k	(3) Crude Net Migr. (per 1k)	(4) IMR (per 1k live births)
<i>Below-median pre-reform unemployment rate</i>				
ATT	-14.23*** (2.24)	-0.331 (0.203)	-5.217 (10.675)	-2.617** (1.146)
Pre-treatment mean	58.72	5.027	0.059	12.332
ATT as % of pre-mean	-24.2%	-6.6%	-8888.1%	-21.2%
<i>N</i> counties	39	32	39	39
<i>Above-median pre-reform unemployment rate</i>				
ATT	-7.37*** (2.13)	-0.454 (0.358)	-11.417 (11.903)	3.213 (2.294)
Pre-treatment mean	66.50	2.550	2.100	12.912
ATT as % of pre-mean	-11.1%	-17.8%	-543.7%	24.9%
<i>N</i> counties	39	27	39	39

Notes: Estimates use the Callaway and Sant’Anna (2021) doubly-robust ATT estimator. Cluster-robust standard errors at the county level are obtained via the multiplier bootstrap with 1000 replications. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, based on two-sided tests with cluster-robust standard errors. Counties are partitioned at the cross-county median of the 1990–1993 mean PR DOL BLS-standard unemployment rate (unemployed / labor force $\times$ 100) (median = 18.62%). The classifier is computed once over the strict pre-rollout window (baseline years = 1990–1993) and is therefore time-invariant within county. counties without a valid baseline value are routed to the below-median bin. *Below-median* = 39 counties; *Above-median* = 39 counties.